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Review

Systematic review of class imbalance problems in manufacturing

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ABSTRACT

Class imbalance (CI) is a well-known problem in data science. Nowadays, it is affecting the data modeling of many of the real-world processes that are being digitized. The manufacturing industry turns out to be highly affected by this problem, especially in fault inspection, prediction or monitoring processes, and in all those processes where the production efficiency is high and the data samples of anomalous events are rare. In this work, we systematically review all the data manipulation, machine learning or deep learning solutions to the CI problem in the manufacturing domain. We also critically evaluate all the different metrics that researchers can compare in order to estimate the improvements carried by their proposed solutions, and we look at the availability of public source code and data-imbalanced datasets that can be used for benchmarking. Finally, we summarize the most applied solutions to the CI problem in manufacturing and we look at future challenges. While posing a reference for the best practices at the time of this review, we challenge researchers to standardize the use of data science algorithms for CI in the manufacturing domain.

1. Introduction

The issue of class imbalance (CI) is well-recognized in the realm of data science [1–3]; however manufacturers might be less aware of its existence and the problems it poses when using machine learning (ML) algorithms. CI presents itself every time that the distribution of positive and negatives samples – of a certain phenomenon that we want to model from the data with ML algorithms – is imbalanced or extremely imbalanced (see Fig. 1). For example, when data about a defective product is collected (negative or minority sample), the amount of these data is extremely less than data about the standard production (positive or majority samples). In terms of six sigma, an efficient production line has few defective parts per million of products within standard parameters. Training a ML classification algorithm without a sufficient amount of balanced samples from majority and minority classes might end up with a classifier that incorrectly aggregates the minority samples into the majority classes, thus not working as expected. This is the CI problem.

The reasoning behind this systematic review of CI problems in the manufacturing domain is that, to our understanding, comprehensive literature reviews dedicated solely to the CI challenge and its resolutions within the manufacturing sector are absent (refer to Section 2). Nonetheless, numerous reviews targeting manufacturing-specific subjects briefly touch upon the CI challenge. For instance, Batool et al. [4] conducted a thorough review on deep learning's role in silicon wafer defect identification, a process notably influenced by skewed class data. In another review centered on big data analytics for smart manufacturing systems, Wang et al. [5] recognized CI as a pivotal concern. Ahmad et al. [6], while surveying deep learning techniques for object spotting in intelligent manufacturing, pinpointed a persisting challenge linked to CI. Similarly, Espadinha-Cruz et al. [7] identified the same challenge during their exploration of data mining applications in semiconductor manufacturing. Li et al. [8] emphasized the difficulty of crafting and assessing tool breakage monitoring (TBM) models in machining due to imbalanced datasets. Guo et al.'s [9] examination of prognostic methods for engineering systems unveiled that many such applications, especially those involving sensor data, grapple with imbalances in data class structures. These reviews, however, generally focus on CI tied to specific topics, not offering holistic solutions for the wider manufacturing arena. We also perused individual articles from niche manufacturing sectors discussing the CI conundrum. Remadna et al. [10], when discussing machine lifespan predictions, highlighted CI concerns. Similarly, Liu et al. [11], while emphasizing the significance of intrusion detection systems for industrial internet of things security, noted the CI challenges in real-world network scenarios. These discussions steered us towards our primary research goal:

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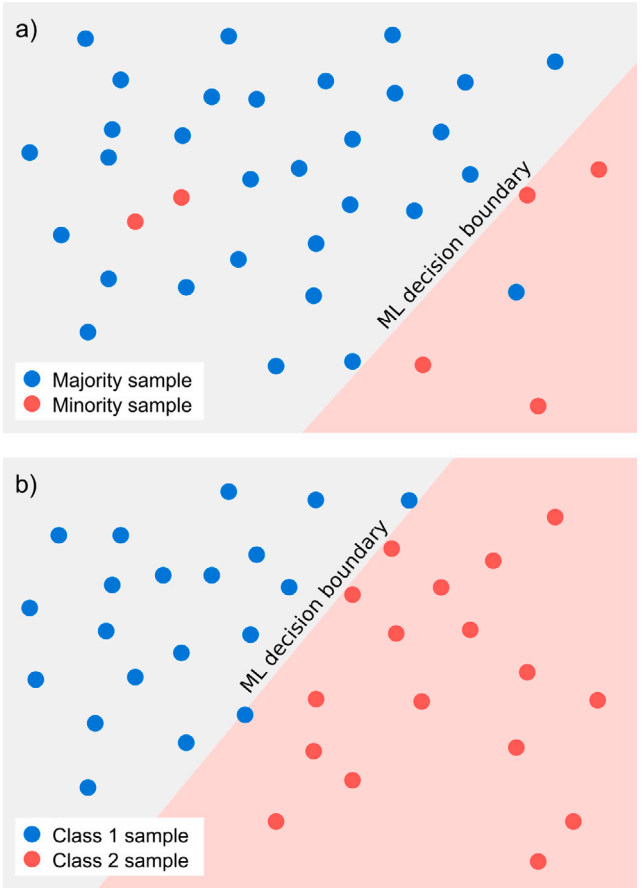


Fig. 1. Class imbalance problem (a): an extreme data imbalance affects the quality of a ML decision boundary to separate the classes, and its solution (b): balanced data allows a ML algorithm to establish a better decision boundary.

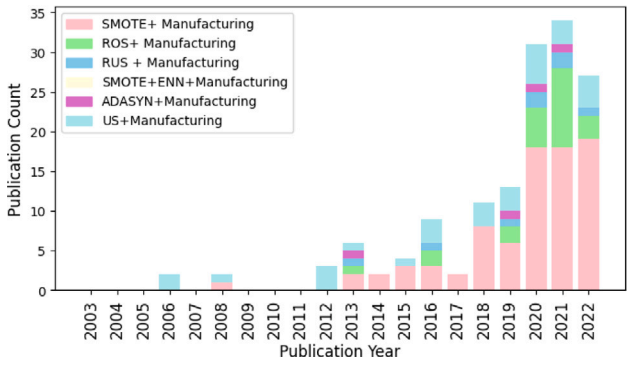


Fig. 2. Progression of data manipulation solutions in manufacturing from scientific literature.

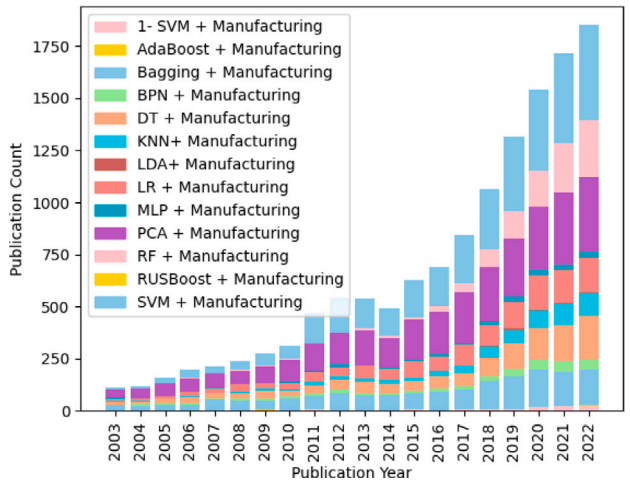


Fig. 3. Progression of machine learning solutions in manufacturing from scientific literature.

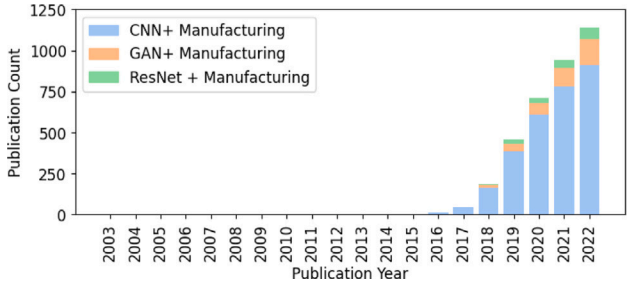


Fig. 4. Progression of deep learning solutions in manufacturing from scientific literature.

comprehensively understanding CI challenges and their remedies in the manufacturing sector.

When considering potential solutions for CI in the literature, numerous scholars, primarily from the computer science domain, have tackled the issue of biased classification in the face of skewed data. They have delved into various enhancement methods. One prominent method revolves around data manipulation (DM), which seeks to augment the data volume to restore balance. Other strategies are rooted in machine learning or deep learning (DL) methodologies. However, distinguishing between conventional ML or DL algorithms and those tailored for DM can be subtle and often presents challenges in differentiation.

From a literature analysis perspective, by looking up to twenty years back, see Figs. 2–4, we can notice how in 2003 ML algorithms were already applied (even though it cannot be said that they were selected to handle CI problems). In 2012, DM algorithms started being cited. Only in 2016, being a more recent advancement, DL started being cited in manufacturing. All the algorithms see an exponential increase during the years. Note that for these charts we specifically looked at the most used DM, ML and DL in manufacturing, as reported in Tables 8–10.

Within the realm of pure Data Manipulation (DM) strategies, data resampling stands out as the predominant method [12,13]; nonetheless, altering the sample count within a dataset can inadvertently inject noise or strip away vital data. Broadly speaking, DM techniques for addressing CI primarily employ various data sampling and re-sampling tactics. These sampling methodologies bifurcate into two primary categories: undersampling and oversampling.

Undersampling strategies involve: randomly reducing samples from the majority class [14], selectively undersampling near the feature space areas adjacent to class boundaries [3], and adopting sequential undersampling, an ensemble approach where multiple models are successively trained. After each pass, accurately identified majority class samples are excised from the training data [15].

Conversely, oversampling techniques encompass strategies such as amplifying the quantity of minority class samples randomly, deliberately oversampling from regions in the feature space proximate to class delineations [3], and fabricating synthetic samples for the minority class [16]. The synthetic creation of new samples has demonstrated enhanced classification results in CI scenarios [17]. Yet, for numerous datasets, especially those from the manufacturing sector, crafting credible synthetic samples remains ambiguous.

One merit undersampling boasts is its computational efficiency during model training, as it demands only a fraction of the majority

class samples. This becomes particularly beneficial in manufacturing settings where datasets often burgeon in size. However, the Achilles' heel of undersampling is the potential data loss from disregarding certain samples. To counteract this, ensemble ML techniques are often employed.

Over the years, ML has ushered in a suite of DM-centric methodologies. For instance, as we venture into data expansion methods, tools such as single-class learning [18–20], ensemble learning [21–23], and cost-sensitive learning [24–26] emerge as potent solutions to the imbalance conundrum.

Among these ML methods, cost-sensitive learning assigns varying misclassification penalties to distinct classes, adeptly navigating the tricky waters of imbalanced data classification. On the other hand, single-class learning zeroes in on training solely the minority class's target samples, significantly bolstering classification efficiency. An illustration of its prowess can be seen in Raskutti and Kowalczyk's work [27], where they demonstrated that one-class support vector machines (1-SVM) outperform their two-class counterparts, especially when CI is stark. A caveat with this method is its susceptibility to overfitting, especially when training revolves exclusively around minority samples, resulting in a dip in generalizability.

Ensemble learning, on the other hand, elevates the classification outcomes for imbalanced data by amalgamating the strengths of multiple classifiers. A shining star in this category is Boosting — an ensemble technique that merges several weaker classifiers to forge a robust one. In this iterative process, each classifier is trained on a data subset, with subsequent models giving precedence to samples that predecessors misclassified. The collective wisdom of these classifiers culminates in the final prediction [17]. While ensemble techniques often yield superior classification outcomes, they come with the baggage of complexity and elongated computation durations.

In contemporary times, Deep Learning (DL) has presented a plethora of effective DM-focused techniques. Notably, generative adversarial networks (GAN) [28,29] and variational autoencoders (VAE) [30] stand out as data generation models. Both GANs and VAEs can leverage limited training samples to produce rudimentary fault samples, even without the backing of extensive datasets. However, these methods demand substantial computational resources. We anticipate finding many of these methodologies being harnessed to address CI issues in manufacturing.

In light of the given premises, our main contribution is to apply the aforementioned classification of CI solutions (i.e., DM, ML and DL approaches) to scientific articles selected in the manufacturing domain and the general aim of this systematic review is to capture an overview of the CI problem in manufacturing by answering the following research questions (RQ):

- **RQ1:** What are the manufacturing operations most affected by CI problems?
- **RQ2:** What are the solutions to CI problems most applied in manufacturing?
- **RQ3:** What are the most used metrics to verify the quality of CI solutions in manufacturing?
- **RQ4:** What is the availability of public code and datasets about CI solutions in manufacturing?

Our article is structured as follows: we report all the related works in Section 2, we present the research methodology for this systematic review in Section 3, we provide an overview of all the findings from our review of the CI problems and solutions in manufacturing in Section 4, we summarize the results about the most used algorithms and metrics, and we report about the publicly available datasets and source code in Section 5, finally, we discuss our results in Section 6, we outline some future challenges in Section 7, and we draw our conclusions in Section 8.

2. Related works

We propose hereby several related works published in the last twenty years – divided in articles (see Section 2.1) and literature reviews (see Section 2.2) – exploring the problem of CI in other fields of research. To the best of our knowledge, none of the related works had a closer view to the CI problem within the manufacturing domain.

2.1. Articles

In 2008, Guo et al. [31] wrote an article exploring the CI problem. In 2009, Chawla [32] made a review of data mining with imbalanced datasets. In 2016, Krawczyk [33] wrote about the future directions and open challenges of learning with imbalanced data, Amin et al. [34] compared the oversampling techniques that can be used to handle the CI problem. In 2017, Ortigosa et al. [35] measured the extent of CI in multi-class problems. In 2020, Thabtah et al. [36] studied the impact of varying CI in classification, Rendon et al. [37] wrote about dealing with the big data multi-class imbalance problem through data sampling methods. In 2022, Tarawneh et al. [38] published an experimental review of more than seventy oversampling methods applied on nine CI datasets and they revealed that all the oversampling methods are prone to generate minority samples that, in reality, belong to the majority class. Standing to their results, oversampling techniques should be considered unreliable for learning from CI data in their current forms and methodologies, thus real-world applications should avoid as much as possible oversampling techniques.

2.2. Literature reviews

In 2007, Mollineda et al. [39] reviewed the CI problem in pattern learning and classification. In 2009, Sun et al. [40] made a review on the classification of imbalanced data, He and Garcia [41] provided a critical review of the research innovations solving CI data. In 2011, Galar et al. [42] reviewed the ensemble algorithms for the CI problem, namely, boosting-, bagging-, and hybrid-based approaches. In 2012, Hoens et al. [43] made an overview of learning from streaming data affected by CI. In 2013, Abd Elrahman and Abraham [44] reviewed the CI problem, Longadge et al. [45] reviewed the CI problem in data mining, Ali et al. [46] made a study on classifications with CI problems, Batuwita and Palade [47] wrote about CI learning methods for support vector machine (SVM), Lin and Chen [48] reviewed the CI classifiers for high-dimensional data. In 2015, Mahmood [49] surveyed the CI learning in data mining, Yu et al. [50] reviewed the CI learning methods in bioinformatics. In 2017, Gosain and Sardana [51] reviewed how to handle the CI problem using oversampling techniques, Shakeel et al. [52] made an exploratory review on the CI problem, Maheshwari et al. [53] reviewed the CI problem and potential solutions, More and Rana [54] reviewed the random forest classification techniques for CI, Haixiang et al. [55] reviewed methods and applications for learning from CI data. In 2018, Leevy et al. [56] surveyed the high-class imbalance in big data, Wang et al. [57] did a systematic study of online CI learning with concept drift, Kaur and Gosain [58] studied the issues and challenges of CI problems in classification, Bi and Zhang [59] made an empirical comparison of the state-of-the-art multi-class imbalance learning algorithms. In 2019, Johnson and Khoshgoftaar [60] surveyed the deep learning articles dealing with CI, Ali et al. [61] reviewed the imbalance class problems in data mining, Ali et al. [62] also reviewed the data preprocessing methods for CI problem, Lanera et al. [63] screened PubMed abstracts to check if CI is always a challenge to ML, Narwane and Sawarkar [64] did a literature survey on ML and CI, Kumar et al. [65] reviewed the ensembles-based approach to overcome the CI problem. In 2020, Devi et al. [66] reviewed the undersampling approaches as a solution to CI problems, Hasib et al. [67] surveyed the methods for managing the classification and solution of CI problems, Oksuz et al. [68] reviewed the CI problems in object detection. In 2021,

Lin et al. [69] reviewed the fuzzy and pattern-based approaches for CI problems, Sampath et al. [70] did a survey in computer vision on the use of generative adversarial networks (GAN) to solve CI problems. In 2022, Santos et al. [71] made a critical review on the joint-effect of CI and overlap, Sharma et al. [72] reviewed the oversampling techniques for CI problems, Sauber-Cole et al. [73] surveyed the use of GAN to alleviate CI in tabular data, Niaz et al. [74] reviewed the ML methods for CI problems and their future challenges.

3. Research methodology

On October 21st, 2022, we searched the Web of Science Core Database for the keywords "manufacturing" and "class imbalance". The search returned 164 results. We exported the results to Rayyan [75], an online tool to aid systematic reviews, and proceeded to eliminate, by abstract review, 70 articles pertaining neither to manufacturing nor CI problems. Thus, 94 articles were initially included into our systematic review. Of these, 20 further articles were excluded, by in depth read, for the following reasons: 17 articles were not in the manufacturing domain, two articles were not accessible, and one was considered poorly written. In order to consider an article in the manufacturing domain, we required that at least one dataset in the manufacturing domain was used to test the CI solution proposed by its authors. We successfully reviewed and included into the results of this systematic review a total of 74 articles. Our systematic review was based on the four research questions presented in the introduction of this article (see Section 1). In order to answer RQ1, we captured details of the manufacturing processes that were either actively discussed, or at least mentioned through a manufacturing-related dataset used to test a CI solution. Regarding RQ2, we went through all the CI solutions proposed in the articles. We present these combined results in Section 4. Finally, we provided answers to RQ3 and RQ4, respectively, by systematically looking at all the metrics, and all the coded solutions and datasets used in the articles. In Section 4, we present an overview of all the solutions that we found. In Section 5, we provide a summary of the results related to our research questions that are finally discussed in Section 6.

4. Overview of class imbalance problems and solutions in manufacturing

We reviewed 74 articles dealing with the CI problem in manufacturing. The solutions that we found are based on data manipulation (DM) (see Table 1), machine learning (ML) (see Tables 2, 3) and/or deep learning (DL) (see Table 4) algorithms. This section contains all the findings, which are also summarized in Table 6. We reported into Tables 1, 2, 3, 4, 6 and 7 all and only the algorithms that are applied in order to reduce the effect of the CI problem. In other words, we filtered out all the algorithms that are accessories to our scope, e.g., the ML classifiers applied to class-balanced data coming from previous algorithms.

Su and Hsiao [151] (see Article 1 in Table 6) implemented a Mahalanobis–Taguchi system (MTS) [207,208] for the inspection process of radio frequencies in mobile phone manufacturing. They compared MTS with a stepwise discriminant analysis (SDA) [223], a C4.5 decision tree (C4.5) [153], a back-propagation neural network (BPN) [150], and a support vector machine (SVM) [227]. MTS, in contrast with the other algorithms, turns out to be quite robust with the imbalanced datasets, even when the training data is highly imbalanced.

Kumar et al. [119] (see Article 2 in Table 6) proposed a framework for fault diagnosis with imbalanced data in cloud-based manufacturing. Firstly, they combined a synthetic minority oversampling technique (SMOTE) [16] (see Section 5.1.5) and a radial basis kernel support vector machine (RBK-SVM) [220] to balance the dataset. Secondly, they trained a logistic regression (LR) [197] classifier. They tested their results on the SPF dataset (see Section 5.5.7).

Table 1
Data manipulation algorithms for class imbalance.

Abbr.	Algorithm	Articles
A-SUWO	Adaptive semi-unsupervised weighted oversampling [76]	[77]
ADASYN	Adaptive synthetic sampling [78]	[77,79–81]
AS	Adaptive sampling (novel) [82]	[82]
AWGN	Additive white Gaussian noise [83]	[84]
Bootstrapping	Bootstrapping [85]	[86]
BSMOTE	Borderline synthetic minority oversampling technique [87]	[77]
BSMOTE-SVM	Borderline SMOTE SVM (novel) [81]	[81]
CADA	Cluster-based adaptive data augmentation (novel) [88]	[88]
COSMOTE	Class ordered synthetic minority oversampling technique (novel) [89]	[89]
CP	Class partitioning [90]	[91]
CSMOTE	Cluster synthetic minority oversampling technique [92]	[77]
ENN	Edited nearest neighbor [93]	[94]
ILR	Imbalanced learning rate (novel) [95]	[95]
IRS	Image resampling (we defined it in Section 5.1.2)	[96–98]
LD	Label dilation [99]	[100]
MGD	Mini-batch gradient descent [101]	[95]
MIP	Morphological image processing [102]	[98]
MM	Memory module [103]	[104]
MWMOTE	Majority weighted minority oversampling technique [105]	[77]
NCL	Neighborhood cleaning rule [106]	[107]
NI	Noise injection [108]	[109]
RM	Random mixing [110]	[109]
RbB	Reliability-based balancing [111]	[112]
ROS	Random oversampling [3]	[77,80,107,113,114]
RUS	Random undersampling [14]	[114–116]
RW	Re-weighting [117]	[114]
SCOTE	Sample-characteristic oversampling technique (novel) [77]	[77]
SIFT-FIM	Scale invariant feature transform features-based image matching [118]	[98]
SMOTE	Synthetic minority oversampling technique [16]	[77,107,112,119–122] [79–81,84,94,113,123]
SMOTE-ENN	SMOTE with ENN [124]	[80,94]
SMOTE-Tomek	SMOTE with Tomek links [125])	[80]
SRA	Sample reliability assessment (novel) [126]	[126]
SSDA	Semi-supervised data augmentation [127]	[100]
TMT	Threshold-moving technique [128]	[112]
UB-SOM	Unstructured borderline self-organizing map (novel) [129]	[129]
US	Undersampling [3,130]	[112,131]
WSMOTE	Weighted synthetic minority oversampling technique [132]	[133]

Tan et al. [112] (see Article 3 in Table 6) combined and compared fuzzy adaptive resonance theory mapping (FAM) neural networks [172], hybrid genetic algorithms (HGA) [183] and dynamic decay adjustment (DDA) [169]. In their approach, both the FAM models and HGA are integrated to classify imbalanced data in semiconductor manufacturing processes. The combinations FAM-DDA, FAM-HGA and FAM-DDA-HGA turned out to be more robust to CI than FAM, HGA or genetic algorithms (GA) [178]. They also compared the FAM-DDA-HGA with the classification results of cost-sensitive artificial neural networks (CS-ANN) [290] trained with data resampling methods such as undersampling (US) [3,130], SMOTE, and threshold-moving technique

Table 2
Machine learning algorithms (1-L) for class imbalance.

Abbr.	Algorithm	Articles
1-SVM	One class support vector (SV) machine [134,135]	[107,136, 137]
A-SVM	SVM with adaptive class weights [138]	[104]
AdaBoost	AdaBoost [139]	[107,140–142] [81,123,143]
AE	Autoencoder [144]	[113]
APH	Axis-parallel hyper-rectangle approach (novel) [145]	[145]
Bagging	Bootstrap aggregating [146]	[107,133, 140]
BGA	Binary genetic algorithm [147]	[148]
BNN-AMSL	Bayesian neural network with additive margin softmax loss [149]	[94]
BPN	Back-propagation neural network [150]	[115,136, 151,152]
C4.5	C4.5 decision tree analysis [153]	[151,152]
C5.0	C5.0 decision tree [154]	[141,155]
CA-PC-SVM	Correlation analysis principal component analysis support vector machine [156]	[157]
CMTSMT2	Complementary fuzzy SVM with SMOTE [158]	[159]
CQ-PC-DT	Chi-square principal component analysis decision tree [160]	[157]
CQ-PC-KNN	Chi-square principal component analysis k-nearest neighbors [161]	[157]
CQ-PC-LR	Chi-square principal component analysis logistics regression [162]	[157]
CQ-PC-NB	Chi-square principal component analysis naïve Bayesian [163]	[157]
CS-SVM	Cost sensitive support vector machine [164]	[107]
DA	Denoising autoencoder [165]	[166]
DAN	Discriminant analysis [167]	[168]
DAMSID	Dyn. AdaBoost.NC with multiple subclassifiers for imb. and drifts (novel) [120]	[120]
DDA	Dynamic decay adjustment [169]	[112]
DEC-SVM	different error cost SVM [47]	[94]
DT	Decision tree [170]	[81,123,143]
DysBoost	Dynamic oversampling approach with adaptive boosting (novel) [171]	[171]
Easy4-j48	Easy ensemble using the j48 decision tree [15]	[159]
FAM	Fuzzy adaptive resonance theory mapping neural networks [172]	[112]
Fast-CBUS-SVM	Fast clustering-based undersampling method using SVM [173]	[159]
FCBNE	Fuzzy class-biased network ensemble (novel) [159]	[159]
FD-KNN	Fault detection using the k-nearest neighbor rule [174]	[107,175]
FFNN	Feed forward sigmoidal neural network [176]	[177]
FVT	Feature-value transformation approach (novel) [145]	[145]
GA	Genetic algorithms [178]	[112]
GA-SVM	GA-optimized multi-class SVM [179]	[122]
GBM	Gradient boost machine [180]	[81,115]
GentleBoost	GentleBoost [181]	[140]
GrC	Granular computing model (novel) [182]	[182]
HGA	Hybrid genetic algorithms [183]	[112]
IC-FDM	Incremental clustering based fault detection method (novel) [184]	[107,184]
IDW-KNN	Inverse distance weights k-nearest neighbor (novel) [185]	[185]
IF	Isolation forest [186]	[137]
IRF	Improved random forest (novel) [142]	[142]
ISVDD-radius-RFE	Imb. SV data description-radius-recursive feature selection (novel) [187]	[187]
KNN	K-nearest neighbor (classifier) [188,189]	[122,123, 152,177]
LDA	Linear discriminant analysis [190,191]	[81,123,187]
Learn++.NIE	Incremental learning of concept drift from imbalanced data [192]	[193]
LightGBM	Light gradient boosting machine [194]	[143]
LOF	Local outlier factor [195]	[137]
LogitBoost	LogitBoost [196]	[140]
LR	Logistic regression [197]	[81,123,177]
LS-SVM	Least squares support vector machine [198]	[77]

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(*continued*).

(TMT) [128]. In addition to that, they also compared FAM-HGA and FAM-DDA-HGA with a reliability-based balancing (RbB) [111] method. In this article, the ML algorithms proposed performed better than the ML algorithms combined with a DM approach used for comparison.

Kwak et al. [184] (see Article 4 in Table 6) proposed an incremental clustering based fault detection method (IC-FDM). The aim of the algorithm is to find wafer faults. It turns to be good even in presence of severe class distribution skews. Their algorithm efficiently processes sensor data by reducing the storage space required. The classification with IC-FDM places the wafer in the closest cluster by using probability-based membership tests and Mahalanobis distance. The algorithm incrementally locates new wafer information in the prototype of the cluster and it merges a few clusters when necessary. Overall, IC-FDM presents high performance in fault detection in environments with stream data.

Lee et al. [107] (see Article 5 in Table 6) compared ML algorithms for CI fault detection problems. They set up two experiments using etching process data and chemical vapor deposition process data, in order to report the performance of four ensemble algorithms, four instance-based algorithms, three sampling-based algorithms, and two SVM algorithms. The three sampling-based algorithms used for DM are SMOTE, neighborhood cleaning rule (NCL) [106] and random oversampling (ROS) [3] (see Section 5.1.3), all followed by either a decision tree (DT) [170] or a SVM classifier. The four ensemble algorithms are bootstrap aggregating (Bagging) [146], AdaBoost [139], SMOTE boost (SBoost) [222] (all followed either by a DT or SVM classifier) and a random forest (RF) [221]. The four instance-based algorithms are fault detection using the k-nearest neighbor rule (FD-KNN) [174], principal component k-nearest neighbor rule (PC-KNN) [217], adaptive Mahalanobis distance k-nearest neighbor rule (MD-KNN) [201] and the IC-FDM from Kwak et al. [184]. The two SVM algorithms are cost sensitive SVM (CS-SVM) [164] and one class SVM (1-SVM) [134,135].

Haddad et al. [82] (see Article 6 in Table 6) presented a multifeature sparse-based defects characterization framework for the automated inspection in the semiconductor manufacturing industry. In order to deal with the CI problem, they proposed and applied a customized version of SMOTE called adaptive sampling (AS). They tested the performance of their approach on a private Intel dataset with real images. As part of the data approach, they augmented with background features subtraction the images, in order to enhance the classification accuracy in presence of a low number of samples.

Ko et al. [148] (see Article 7 in Table 6) proposed a binary genetic algorithm (BGA) [147] based wrapper to deal with CI, followed by several machine learning anomaly detection algorithms, for quality monitoring in manufacturing. They used data from quality inspection and after-sales service.

Table 3
Machine learning algorithms (M-Z) for class imbalance.

Abbr.	Algorithm	Articles
MBDT	Main branch decision tree (novel) [199]	[199]
MCMND	Modified classifier for matrix normal distributions (novel) [177]	[177]
MD	Mahalanobis distance [200]	[168]
MD-KNN	Adaptive Mahalanobis distance k-nearest neighbor rule [201]	[107]
MGA	Mahalanobis genetic algorithm (novel) [121]	[121]
MLP	Multi-layer perceptron [202]	[81,123,142,159]
MLP-MFE	Multi-layer perceptron with mean false error [203]	[94]
MLP-MSFE	Multi-layer perceptron with mean squared false error [203]	[94]
MMTS	Multiclass Mahalanobis–Taguchi system [204]	[205]
MPCA-D	Multiway principal component analysis D-statistics [206]	[166]
MPCA-Q	Multiway principal component analysis Q-statistics [206]	[166]
MTS	Mahalanobis–Taguchi system [207,208]	[121,151]
NB	Naïve Bayes [172]	[81,177]
NNE	Neural networks ensembles [209]	[116]
OCSNN	Online cost-sensitive neural network [210,211]	[212]
PARAFAC	Parallel factor analysis [213]	[166]
PC-ANN	Principal component (PC) analysis artificial neural network [214]	[157]
PCA	PC analysis [215]	[123,157,187]
PC-BDT	PC analysis boosted tree C4.5 [216]	[157]
PC-KNN	PC k-nearest neighbor rule [217]	[107]
PC-SD-KNN	Total squared distance of k-nearest neighbor in PC space [161]	[166]
PC-SPE	PC analysis with squared prediction error [218]	[175]
PC-T2	PC analysis with Hotelling's T^2 [218]	[175]
PLS	Partial least squares regression [219]	[115]
RBK-SVM	Radial basis kernel support vector machine [220]	[119,159]
RF	Random forest [221]	[91,107,122,141,143] [81,123,152,221]
RUSBoost	Random undersampling boost [23]	[116,143,193]
SBoost	Synthetic minority oversampling technique boost [222]	[107]
SD-KNN	Total squared distance of k-nearest neighbor [174]	[166]
SDA	Stepwise discriminant analysis [223]	[151]
SLWE	Smooth local weighted edge (novel) [212]	[212]
SMFA	Sensitive marginal Fisher analysis [224]	[133]
Softmax	Softmax (classifier) [225]	[104]
SVDD	Support vector data description [226]	[175,187]
SVM	Support vector machine [227]	[81,136,142,151,152,227]
SVR	Support vector regression [228]	[115]
XGBoost	Extreme gradient boosting [229]	[81,143]
Z-SVM	Z-SVM [230]	[94]

Saqlain et al. [96] (see Article 8 in Table 6) proposed a deep learning-based convolutional neural networks (CNN) [245] for automatic wafer defect identification in the semiconductor manufacturing process. They applied data augmentation using image resampling (IRS) methods consisting of width shift, horizontal flipping, height shift, random rotation of 10-degree, channel shifting, zooming of the input images and shearing range. The method was evaluated on the public WM811K dataset [289] of wafer defect maps (see Section 5.5.11).

Table 4
Deep learning algorithms for class imbalance.

Abbr.	Algorithm	Articles
3D-CNN	3D convolutional neural network [231]	[232]
3D-GAN	3D generative adversarial network (novel) [232]	[232]
3D ResNet-34	3D-ResNet-34 [233]	[232]
ACB-GCN	Anchor-based class-balanced graph convolutional network (novel) [234]	[234]
ACGAN	Auxiliary classifier generative adversarial network [235]	[236]
Balanced-YOLOv3	Balanced-YOLOv3 (novel) [237]	[237]
C3D	3D convolutional neural network [231]	[232]
CAE	Convolutional autoencoder [238]	[236]
Camem-BERT	CamemBERT [239]	[240]
CGAN	Conditional generative adversarial networks [241]	[142]
CIATL	Class-imbalance adversarial transfer learning (novel) [242]	[242]
CLSTM	Convolutional long short-term memory [243]	[244]
CNN	Convolutional neural network [245]	[96–98,100,136] [95,104,114,152]
CNN-AD	Convolutional neural network for abnormal detection (novel) [136]	[136]
CSM-VIT	Contrastive-weighted self-supervised model with VIT (novel) [114]	[114]
CVAE	Conditional variational autoencoder [246]	[236]
DL-BCE-Dice	DL binary cross-entropy [247] Dice [248] loss functions (novel) [249]	[249]
DNN	Deep neural network [250]	[251]
DNN-Ens	Ensembles of DNN [252]	[251]
FGN	Fault feature generation (novel) [253]	[253]
GAN	Generative adversarial network [28,29]	[80,113,254–256]
GCForest	Multi-grained cascade forest [257]	[152]
GoogleNet	GoogleNet [258]	[152]
GPT-2	Language model GPT-2 [259]	[240]
ICNN	Improved convolutional neural network (novel) [126]	[126]
ImageNet	ImageNet [260]	[152]
KNN-CNN	K-nearest neighbor convolutional neural network [261]	[104]
LMAN	Latent mapping adversarial network (novel) [256]	[256]
MGGAN	Multigranularity generative adversarial network (novel) [262]	[262]
MM-CNN	Memory module augmented convolutional neural network (novel) [104]	[104]
MoCo	Momentum contrastive network [263]	[114]
MoViNet-A2	MoViNet-A2 [264]	[232]
PSO-DBN	Particle swarm optimization deep belief network (novel) [157]	[157]
ResNet	Residual network [265]	[114,152]
SimCLR	Advanced contrastive learning method [266]	[114]
Softmax-CNN	Softmax convolutional neural network [267]	[104]
SSCN	Self-supervised convolutional network [268]	[114]

(continued on next page)

Table 4 (*continued*).

Abbr.	Algorithm	Articles
STA	Spatio-temporal autoencoder (STA) [269]	[232]
TR-FL	Transformer [270] with focal loss [271]	[272]
VAE	Variational autoencoder [273,274]	[79]
VIT	Augmented vision transformer [275]	[114]
WELM	Weighted extreme learning machine [276]	[244]
WGAN	Wasserstein general additional networks [277]	[113]
WGAN-GP	Wasserstein generative adversarial with gradient penalty network [278]	[244]

Table 5
Public manufacturing datasets with class imbalance.

Abbr.	Dataset	CI ratio	Articles
CRWU bearing fault	Case Western Reserve University bearing fault [279]	Unspecified	[77,126, 244,253]
GC10-DET	GC10-DET manufacturing defects [280]	Unspecified	[234]
GDXray	GDXray (castings, welds, baggage, nature objects, and settings) [281]	Unspecified	[249]
NEU-CLS-64	NEU-CLS-64 manufacturing defects [282]	Unspecified	[234]
SECOM	Semiconductor manufacturing process [283]	1:14	[81,123, 157,187] [80,129, 187]
Severstal	Severstal steel manufacturing process [284]	Unspecified	[256]
SPF	Steel plates faults from UCI machine learning repository [285]	Multiple ratios	[254,255]
SW	Semiconductor wafer [286]	10.7%	[113]
TD	Textile defects [287]	Unspecified	[234]
WD	Wood defects [288]	Unspecified	[234]
WM811K	Wafer maps [289]	Unspecified	[96,109, 133,152, 262]

Lin et al. [120] (see Article 9 in Table 6) proposed a dynamic AdaBoost.NC with multiple subclassifiers for imbalance and drifts (DAM-SID) for condition-based maintenance. They applied SMOTE to the imbalanced class data in their experimental dataset.

El-Banna [121] (see Article 10 in Table 6) proposed a Mahalanobis genetic algorithm (MGA) classifier to address the problem of feature selection for imbalanced spot welding data. The classifier was evaluated in a real-life dataset from the spot welding process and compared with an MTS, other than with and without applying SMOTE as an additional data manipulation step.

Park et al. [97] (see Article 11 in Table 6) proposed a solution for welding defects automatic inspection based on CNN. Before using the CNN, they generated additional training samples with IRS methods, i.e. by randomly transforming and distorting raw image samples, thus performing a form of data augmentation applicable to image data. The model was experimentally evaluated.

Xu et al. [100] (see Article 12 in Table 6) proposed to perform defect inspection with a small data-driven CNN. The CI problem was tackled with two methods, namely, label dilation (LD) [99] and semi-supervised data augmentation (SSDA) [127]. The CNN architectures compared were Inception v3 [291], VGG-16 [292], ResNet-18 [265] and SqueezeNet v1.1 [293]. Their performance was evaluated on a roller bearing manufacturing dataset.

Du [175] (see Article 13 in Table 6) applied and compared several one-class classifiers for imbalanced data, namely, principal component analysis with Hotelling's T^2 (PC-T2) [218], principal component analysis with squared prediction error (PC-SPE) [218], FD-KNN and support vector data description (SVDD) [226]. The anomaly detection was performed on real semiconductor etching process data by extracting bispectrum and PCA-based features. In their experiment, the features of bispectrum and PCA were found to have very different values for the normal and faulty classes, thus, they provided an accurate and robust method to discriminate between the two classes.

Chen and Su [182] (see Article 14 in Table 6) introduced a novel knowledge acquisition algorithm called granular computing model (GrC) for imbalanced data based on FAM. They tested it on a scheduling task in a flexible manufacturing system environment simulated with imbalanced data.

Han et al. [140] (see Article 15 in Table 6) compared four ensemble algorithms consisting of Bagging and three boosting algorithms, namely, AdaBoost, LogitBoost [196] and GentleBoost [181], for the quality monitoring of planetary gear packs.

Zuvela et al. [141] (see Article 16 in Table 6) compared three ensemble methods for quality monitoring of an industrial process, namely, C5.0 decision tree (C5.0) [154], AdaBoost and RF. They tested them on a real industrial dataset acquired from the inspection of surface waviness in the glass substrates of flat panel displays, which was affected by CI.

Zheng et al. [142] (see Article 17 in Table 6) researched an efficient and easily implementable rescheduling strategy for the remanufacturing of aircraft parts. They applied a DL conditional generative adversarial networks (CGAN) [241] for data augmentation, before using a ML classifier. They also proposed an improved random forest (IRF) as classifier, and compared it with AdaBoost, multi-layer perceptron (MLP) [202] and SVM. Their results indicated that CGAN can improve the geometric mean (G-Mean) (see Section 5.4.6) by 3.5%–5% with all the tested classifiers.

Kim et al. [157] (see Article 18 in Table 6) proposed a particle swarm optimization deep belief network PSO-DBN algorithm for quality monitoring with CI. The data was firstly oversampled with SMOTE. Their classifier was evaluated on the SECOM dataset (see Section 5.5.5), together with other algorithms, namely, principal component analysis (PCA) [215] artificial neural network (PC-ANN) [214], PCA boosted tree C4.5 (PC-BDT) [216], chi-square (CQ) [294] PCA k-nearest neighbors (KNN) [189] (CQ-PC-KNN) [161], CQ PCA naïve Bayesian (CQ-PC-NB) [163], CQ PCA logistics regression (CQ-PC-LR) [162], CQ PCA DT (CQ-PC-DT) [160] and correlation analysis (CA) [295] PCA SVM (CA-PC-SVM) [156].

Wei et al. [77] (see Article 19 in Table 6) proposed a new DM method called sample-characteristic oversampling technique (SCOTE) to improve bearing fault diagnosis with a multi-class least squares support vector machine (LS-SVM) [198]. They evaluated their method on the CRWU bearing fault dataset (see Section 5.5.1). SCOTE was compared, in conjunction with LS-SVM, with other DM algorithms, namely, ROS, SMOTE, borderline SMOTE (BSMOTE) [87], cluster SMOTE (CSMOTE) [92], majority weighted minority oversampling technique (MWMOTE) [105], adaptive semi-unsupervised weighted oversampling (A-SUWO) [76] and adaptive synthetic sampling (ADASYN) [78].

Jang et al. [166] (see Article 20 in Table 6) compared a denoising autoencoder (DA) [165], through a fault monitoring index, with five one-class multivariate fault detection algorithms, namely, multiway principal component analysis D-statistics (MPCA-D) [206], multiway principal component analysis Q-statistics (MPCA-Q) [206], parallel factor analysis (PARAFAC) [213], total squared distance of KNN (SD-KNN) [174] and SD-KNN in principal component space (PC-SD-KNN) [161]. Their study was aimed at classifying faulty wafers in the semiconductor manufacturing process in presence of CI.

Table 6
Summary of findings (Articles 1–37).

Art.	Ref.	Approach	Algorithms
1	[151]	ML	BPN, C4.5, MTS (best), SDA, SVM
2	[119]	DM + ML	SMOTE + RBK-SVM
3	[112]	DM + ML	RbB, SMOTE, TMT, US + DDA, FAM, FAM-DDA, FAM-DDA-HGA (best), FAM-HGA, GA, HGA
4	[184]	ML	IC-FDM (novel)
5	[107]	DM + ML	NCL, ROS, SMOTE + 1-SVM, AdaBoost, Bagging, CS-SVM, FD-KNN, IC-FDM, MD-kNN, PC-KNN, RF, SBoost
6	[82]	DM	AS
7	[148]	ML	BGA
8	[96]	DM + DL	IRS + CNN
9	[120]	DM + ML	SMOTE + DAMSID
10	[121]	DM + ML	MGA, MTS, SMOTE-MGA (best), SMOTE-MTS
11	[97]	DM + DL	IRS + CNN
12	[100]	DM + DL	LD, SSDA + CNN (SqueezeNet v1.1, Inception v3, VGG-16, ResNet-18)
13	[175]	ML	FD-KNN (best), PC-SPE, PC-T2, SVDD (best)
14	[182]	ML	GrC
15	[140]	ML	AdaBoost, Bagging, GentleBoost, LogitBoost (best)
16	[141]	ML	AdaBoost, C5.0, RF (best)
17	[142]	DL + ML	CGAN + AdaBoost, IRF (best), MLP, SVM
18	[157]	DM + ML-DL	SMOTE + CA-PC-SVM, CQ-PC-DT, CQ-PC-KNN, CQ-PC-LR, CQ-PC-NB, PC-ANN, PCA, PC-BDT, PSO-DBN
19	[77]	DM + ML	A-SUWO, ADASYN, BSMOTE, CSMOTE, MWMOTE, ROS, SCOTE, SMOTE + LS-SVM
20	[166]	ML	DA, MPCA-D, MPCA-Q, PARAFAC, PC-SD-KNN, SD-KNN
21	[133]	DM + ML	WSMOTE + Bagging, SMFA
22	[136]	ML-DL	1-SVM, BPN, CNN, CNN-AD (best), SVM
23	[98]	DM + DL	IRS, MIP, SIFT-FIM + CNN (ShuffleNet V2)
24	[115]	DM + ML	RUS + BPN, GBM, PLS, SVR
25	[104]	DM + DL	A-SVM, KNN-CNN, MM-CNN (novel), Softmax-CNN
26	[122]	DM + ML	SMOTE + GA-SVM, KNN, RF
27	[253]	DL	BPN, CNN, FGN (novel), KNN, RF (best), SVM
28	[143]	ML	AdaBoost, LightGBM, RF, RUSBoost-DT, XGBoost
29	[86]	DM	Bootstrapping
30	[187]	ML	ISVDD-radius-RFE, LDA, PCA, SVDD
31	[95]	DM + DL	ILR (novel), MGD + CNN
32	[193]	ML	Learn++.NIE, RUSBoost (batch vs online)
33	[109]	DM	NI, RM
34	[91]	DM + ML	CP + RF
35	[199]	ML	MBDT (novel)
36	[254]	ML	GAN
37	[177]	ML	FFNN (worse), KNN, LR, MCMND (novel), NB (worse), RF, SVM

Wang et al. [133] (see Article 21 in Table 6) proposed a new fault detection method named ensemble manifold sensitive margin fisher analysis (EMSMFA) for the semiconductor manufacturing process. EMSMFA is based on Bagging, working together with weighted synthetic minority oversampling technique (WSMOTE) [132] and sensitive marginal Fisher analysis (SMFA) [224]. Their work was evaluated on the SECOM dataset (see Section 5.5.5).

Li et al. [136] (see Article 22 in Table 6) proposed a convolutional neural network for abnormal detection (CNN-AD) in tool breakage prediction aimed at solving the CI problem. They made a comparison between their architecture and CNN, BPN, SVM and 1-SVM.

Liang et al. [98] (see Article 23 in Table 6) used a ShuffleNet V2 [296] CNN architecture to identify ink-jet codes on complex backgrounds. They applied image processing methods to deal with data imbalance, namely, IRS, morphological image processing (MIP) [102], and scale invariant feature transform features-based image matching (SIFT-FIM) [118]. IRS methods included region translation, random patch cropping, small angle image rotation, horizontal flipping, random contrast change, random brightness change and image resizing.

Hung et al. [115] (see Article 24 in Table 6) studied a solution for the automatic identification of delamination in the semiconductor assembly process. They pre-processed the imbalanced data with random undersampling (RUS) [14] and compared four ML architectures, namely, gradient boost machine (GBM) [180], BPN, support vector regression (SVR) [228], and partial least squares regression (PLS) [219]. They tested their approach on a dataset obtained from a leading Taiwanese semiconductor assembly manufacturer.

Hyun and Kim [104] (see Article 25 in Table 6) proposed a memory-module augmented CNN (MM-CNN) with triplet loss for classifying defect patterns in highly CI wafer bin map data, where a memory module (MM) [103] is adopted to balance the data of defect-pattern classes. During the training, the same memory size is allocated for each class, in order to prevent samples belonging to the majority class from occupying more memory than the minority class samples. This method balances the prediction data and improves the classification accuracy. They compared their proposed model with a CNN where the data imbalance is treated with Softmax [225] (Softmax-CNN [267]) and KNN [188] (KNN-CNN) [261] classifiers. They also compared it with a SVM with adaptive class weights (A-SVM) [138].

Hui et al. [122] (see Article 26 in Table 6) proposed a normalized mutual information and random sampling with replacement variable selection method in assembly quality monitoring, followed by SMOTE to deal with the CI problem, and GA-optimized multi-class SVM (GA-SVM) [179] as a classifier. They used k-nearest neighbors (KNN) [188, 189] and RF as alternative classifiers for comparison.

Table 7
Summary of findings (Articles 38–74).

Art.	Ref.	Approach	Algorithms
38	[131]	DM	US
39	[137]	ML	1-SVM, IF, LOF
40	[113]	DM + ML + DL	ROS, SMOTE + AE + GAN, WGAN
41	[212]	ML	OCSNN, SLWE (novel)
42	[79]	DM + ML	ADASYN, SMOTE + VAE
43	[116]	DM + ML	RUS + NNE, RUSBoost
44	[152]	ML + DL	BPN, C4.5, KNN, RF, SVM + CNN (GoogleNet, ImageNet, ResNet), GCForest, DenseNet-GCF
45	[123]	DM + ML	SMOTE + AdaBoost, DT, KNN, LDA, LR, MLP, PCA, RF
46	[234]	DL	ACB-GCN (novel)
47	[94]	DM + ML	ENN, SMOTE, SMOTE-ENN + BNN-AMSL (best), DEC-SVM, MLP-MFE, MLP-MSFE, Z-SVM
48	[159]	DM + ML	CMTSMT2, Easy4-j48, Fast-CBUS-SVM, FCBNE (novel), MLP, RBK-SVM
49	[168]	ML	DAN, MD (best)
50	[155]	ML	C5.0
51	[255]	DL	GAN
52	[145]	DM + ML	APH (novel), FVT (novel)
53	[262]	DL	MGGAN (novel)
54	[232]	DL	3D-CNN, 3D-GAN (novel), 3D-ResNet-34, C3D, MoViNet-A2
55	[272]	DM + DL	TR-FL
56	[84]	DM	AWGN, SMOTE
57	[171]	ML	DysBoost (novel)
58	[126]	DM + DL	SRA (novel) + ICNN (novel)
59	[89]	DM	COSMOTE (novel)
60	[249]	DL	DL-BCE-Dice (novel)
61	[240]	DL	GPT-2/CamemBERT
62	[237]	DL	Balanced-YOLOv3 (novel)
63	[242]	DL	CIATL (novel)
64	[88]	DM	CADA (novel)
65	[114]	DM + DL	ROS, RUS, RW + CSM-VIT (novel), MoCo, CNN (ResNet), SimCLR, SSCN
66	[80]	DM + DL	ADASYN, ROS, SMOTE, SMOTE-ENN, SMOTE-Tomek + GAN
67	[256]	DL	GAN, LMAN (novel)
68	[251]	DL	DNN-Ens
69	[244]	DL	CLSTM, WELM, WGAN-GP
70	[205]	ML	MMTS
71	[81]	DM + ML	ADASYN, BSMOTE-SVM (novel), SMOTE + AdaBoost, DT, GBM, LDA, LR, MLP, NB, RF, SVM, XGBoost
72	[185]	ML	IDW-KNN (novel)
73	[129]	DM	UB-SOM (novel)
74	[236]	DL	ACGAN, CAE, CVAE

Pan et al. [253] (see Article 27 in Table 6) focused on rolling bearing zero-shot fault detection, an extreme case of CI. They proposed a fault feature generation (FGN) deep learning algorithm inspired by CGAN made of a generator, a discriminator, and an additional feature extractor. The latter element extracts features from the vibration signals and reduces the data dimension. FGN is used to carry out clustering operations, other than recognizing the abnormal samples. FGN was compared with other architectures, such as RF, SVM, KNN, BPN and CNN. Their architecture was evaluated on three datasets, namely, the CRWU bearing fault dataset (see Section 5.5.1), other than a privately acquired dataset and another CI dataset in electronics. By means of accuracy, RF seems to outperform both FGN and the other architectures on all three datasets.

Liu et al. [143] (see Article 28 in Table 6) proposed to use decision tree (DT) [170] with random undersampling boosting (RUSBoost) [23] (see Section 5.2.12) in the quality monitoring of a battery manufacturing process. The ensemble learning framework was proposed to compensate for the CI problem. Their solution was compared with RF, extreme gradient boosting (XGBoost) [229], light gradient boosting machine (LightGBM) [194] and AdaBoost.

Yang et al. [86] (see Article 29 in Table 6) opted for an oversampling of the minority class data, with the resampling implemented via Bootstrapping [85] for quality monitoring in the rail manufacturing process.

Wang et al. [187] (see Article 30 in Table 6) proposed a novel algorithm called imbalanced support vector data description-radius-recursive feature selection (ISVDD-radius-RFE). They used two datasets, the SECOM dataset (see Section 5.5.5) in manufacturing, and one for general machine learning. Their architecture was compared with PCA, linear discriminant analysis (LDA) [190,191] and SVDD.

He et al. [95] (see Article 31 in Table 6) applied two techniques to fix the imbalance class problem in the semiconductor manufacturing process. Firstly, they applied a mini-batch gradient descent (MGD) [101] to regroup the training samples. Secondly, they proposed and applied a novel method called the imbalanced learning rate (ILR). The rebalanced dataset was then used with a CNN for classification. Their results were evaluated on public data-imbalanced datasets that are not in the manufacturing domain.

Chen and Boning [193] (see Article 32 in Table 6) applied two versions of RUSBoost (see Section 5.2.12), the first one in batch mode, and the second one in online mode, for quality monitoring in the semiconductor manufacturing processes. Their results were compared with the incremental learning of concept drift from CI data (Learn++.NIE) [192] and tested on privately acquired datasets.

di Bella et al. [109] (see Article 33 in Table 6) worked on a data augmentation procedure designed for Wafer Defect Maps (WDM). In particular, they performed two transformations called noise injection

(NI) [108] and random mixing (RM) [110]. They tested their solutions on the public WM811K dataset [289] of wafer defect maps (see Section 5.5.11), and on a private dataset acquired from a production site.

Hirsch et al. [91] (see Article 34 in Table 6) dealt with the multiclass imbalance problem in manufacturing quality monitoring. They used a class partitioning (CP) technique [90] to generate minority classes and majority classes subsets and trained different ensembles of RF on each subset. They generated a ranked list of predicted classes. The private dataset used contains 1,050 samples assigned to 84 different classes.

Hsu et al. [199] (see Article 35 in Table 6) proposed and directly applied a ML algorithm, namely, main branch decision tree (MBDT), to an imbalanced dataset of semiconductor manufacturing process yield to identify the root causes of low yield (minority class).

Lee and Baek [254] (see Article 36 in Table 6) used a generative adversarial network (GAN) [28,29] (see Section 5.3.2) to augment the dataset with fault (minority class) samples. They tested their GAN on the SPF dataset (see Section 5.5.7).

Rafajlowicz [177] (see Article 37 in Table 6) introduced a modified classifier for matrix normal distributions (MCMND) and compared it with a logistic regression (LR) classifier with an L2 regularization coefficient equal to one [197], a naïve Bayes (NB) [172] classifier, a feed forward sigmoidal neural network (FFNN) [176] classifier, a RF classifier, a KNN classifier, and a SVM classifier. The article demonstrated that when feeding raw images to the classifiers, these can directly handle CI through their implicit robustness. Among their results, all of the above algorithms perform well enough, except for NB and FFNN classifiers.

Haddad et al. [131] (see Article 38 in Table 7) proposed an ad-hoc classifier for quality control in the semiconductor manufacturing processes. They solved the imbalance class problem with US (see Section 5.1.7) and oversampling [130] techniques. The ML classifiers compared were not analyzed for their ability to handle CI problems.

Zeng et al. [137] (see Article 39 in Table 7) applied isolation forest (IF) [186] and compared it with two other anomaly detection algorithms, namely 1-SVM and local outlier factor (LOF) [195].

Song and Baek [113] (see Article 40 in Table 7) proposed the autoencoder Wasserstein general advertising networks (AE-WGAN). Firstly, they trained an autoencoder (AE) [144] with normal data. Secondly, they oversampled the abnormal data using Wasserstein general additional networks (WGAN) [277], until the number of instances in the normal and abnormal data were equal. Thirdly, they carried out an efficient anomaly detection through the previously learned AE. They compared the results of AEWGAN with random oversampling (ROS) [3] (see Section 5.1.3), synthetic minority oversampling technique (SMOTE) [16] (see Section 5.1.5) and GAN (see Section 5.3.2). They used the SW dataset (see Section 5.5.8). They conducted experiments using only 30 abnormal data samples of the 762 present in the dataset, corresponding to an imbalance ratio of 0.5%, in order to equiparate the CI data ratio to the 3-sigma of the actual process.

Lee [212] (see Article 41 in Table 7) proposed a smooth local weighted edge (SLWE) machine learning algorithm for imbalanced class data and compared it with an online cost-sensitive neural network (OCSNN) [210,211]. They evaluated their method using 63 sets of unpublished data from a semiconductor company corresponding to different production batches. Each production batch contains from 2,280 to 4,922 samples, with an imbalance ratio of 15.41%.

Lee et al. [79] (see Article 42 in Table 7) used oversampling methods such as SMOTE [16] (see Section 5.1.5) and ADASYN [78] (see Section 5.1.1), other than a variational autoencoder (VAE) [273,274], which can be applied as ML data augmentation method, to generate synthetic data with sufficient variance.

Cinar et al. [116] (see Article 43 in Table 7) looked at automated quality inspection methods in the printed circuit board industry in presence of class imbalanced data. They applied algorithmic solutions based on undersampling methods such as RUSBoost [23] (see Section 5.2.12) and neural networks ensembles (NNE) [209] trained with fixed or randomly varied topologies and feature sets. Although the ensembles of neural networks had a slightly better performance than RUSBoost, the data undersampling technique applied was the same one: random undersampling (RUS) [14] (see Section 5.1.4).

Yu et al. [152] (see Article 44 in Table 7) worked on a wafer map defect recognition deep learning algorithm called densely connected convolutional network — deep forest (DenseNet-GCF). Firstly, they transferred the network weight parameters of ImageNet [260] to a CNN that became a densely connected convolutional network (DenseNet) [297]. Then, they redesigned the classification layer adding a deep forest, also known as multi-grained cascade forest (GCForest) [257], in order to identify the wafer defect pattern based on the abstract features from the wafer maps extracted by DenseNet. DenseNet-GCF results more robust to CI than the deep convolutional neural network GoogleNet [258], residual network (ResNet) [265], GCForest, RF, SVM with linear kernel function and Gaussian kernel function, KNN and C4.5. Also, the choice of CGForest for the classification layer beats a choice of BPN, RF, SVM, KNN and C4.5 as alternative classification layers. They tested their results on the WM811K dataset (see Section 5.5.11).

Chazhoor et al. [123] (see Article 45 in Table 7) tested a few dimensionality reduction methods such as PCA and LDA, in conjunction with ML algorithms such as RF, MLP, LR, AdaBoost and DT. They also tried with SMOTE followed by RF, MLP, LR and KNN. SMOTE performed best in terms of accuracy. For their tests, they used the semiconductor manufacturing process (SECOM) dataset (see Section 5.5.5).

Wang et al. [234] (see Article 46 in Table 7) proposed an anchor-based class-balanced graph convolutional network (ACB-GCN) deep learning algorithm for surface defects recognition. ACB-GCN is built to overcome the coupling between CI, intraclass variations of surface defects and interclass similarities. Since excessive information from majority classes influences the performance of graph convolution, their method constructs a class-balanced graph to address that problem, and then it defines anchor vectors in each defect to reduce the influence of interclass similarities and intraclass variations. They performed experiments on four public datasets with CI, namely, NEU-CLS-64 (see Section 5.5.4), the WD dataset (see Section 5.5.10), the TD dataset (see Section 5.5.9) and GC10-DET for large-scale metallic surface defect detection (see Section 5.5.2).

Park et al. [94] (see Article 47 in Table 7) tackled the problem of predicting semiconductor chip-level defects with a Bayesian neural network (BNN) [298] with additive margin softmax loss (BNN-AMSL) [149], which incorporates sample- and class-level uncertainties, and their relative frequencies. In order to prove the robustness of their BNN-AMSL algorithm, they used two sets of rebalanced data with imbalance ratios of 10:1 and 30:1, other than the original dataset with imbalance ratio of 16:1, and compared their method with other available methods. For the CI learning methods, they considered two approaches: resampling and cost-sensitive methods. For the resampling methods, they considered the edited nearest neighbor (ENN) [93], SMOTE, and SMOTE combined with ENN (SMOTE-ENN) [124]. These re-sampling methods were combined with a classifier, namely a multilayer perceptron (MLP) [202]. For the cost-sensitive methods, they considered a MLP combined with two loss functions to solve the imbalance, namely, mean squared false error (MLP-MSFE) [203] and mean false error (MLP-MFE) [203], and two support vector machine (SVM)-based methods [227], namely, Z-SVM [230] (a SVM for improved classification of imbalanced data) and a different error cost SVM (DEC-SVM) [47]. In all cases, their proposed algorithm performed better.

Pillai et al. [159] (see Article 48 in Table 7) proposed a fuzzy class-biased network ensemble (FCBNE). The performance of the FCBNE is compared on an industrial dataset with five benchmark models: RBK-SVM, MLP with three fully connected layers, a complementary fuzzy

SVM with SMOTE (CMTSMT2) [158], two undersampling methods, namely, an easy ensemble (Easy4) using the j48 decision tree classifier [15] (Easy4-j48) and a fast clustering-based undersampling method (Fast-CBUS) [173] using SVM as a classifier.

Yang et al. [168] (see Article 49 in Table 7) compared two machine learning algorithms, namely, the Mahalanobis distance (MD) [200] and the discriminant analysis (DAN) [167], on a color filter manufacturing quality process. Their results show that MD performs better than DA, especially when the imbalance ratio grows.

Jazi and Liu [155] (see Article 50 in Table 7) used C5.0, a cost-sensitive decision tree classifier, for the quality inspection in the thin-film transistor liquid crystal displays manufacturing process.

Lee et al. [255] (see Article 51 in Table 7) proposed a generative-rehearsal strategy using a GAN for incremental learning with unbalanced data. They used the SPF dataset (see Section 5.5.7).

Graur et al. [145] (see Article 52 in Table 7) looked at automatic product inspection machines based on the multiple instance problem in which an entity may be faulty due to one or more components. Furthermore, the faulty components belong to a minority class. They proposed a feature-value transformation (FVT) DM/ML approach and an axis-parallel hyper-rectangle (APH) DM/ML approach inspired by the iterated discrimination algorithm (IDA) [299].

Yu and Liu [262] (see Article 53 in Table 7) proposed a multigranularity generative adversarial network (MGGAN) for the recognition of wafer map defects. They tested their solutions on the public WM811K dataset [289] of wafer defect maps (see Section 5.5.11).

da Silva et al. [232] (see Article 54 in Table 7) compared six different deep learning models for defect detection: a proposed 3D generative adversarial network (3D-GAN), a custom 3D convolutional neural network (3D-CNN) [231], a spatio-temporal autoencoder [269], a 3D ResNet-34 [233], a 3D convolutional neural network (C3D) [231], and a MoViNet-A2 [264]. They introduced three novel video datasets that simulate real-world industry-oriented failures, namely, 60frames, DenserGlitch and The_1R, however, none of them was published.

Usuga-Cadavid et al. [272] (see Article 55 in Table 7) used a transformer (TR) [270], a deep learning algorithm, for harnessing data from historical maintenance datasets. In order to tackle the CI problem, they applied a loss function called focal loss (FL) [271]. The resulting architecture is hereby classified as TR-FL.

Martins et al. [84] (see Article 56 in Table 7) proposed to use a hybrid method of data augmentation, comprising both an additive white Gaussian noise (AWGN) [83] and a synthetic minority oversampling technique (SMOTE) [16].

Qin et al. [171] (see Article 57 in Table 7) proposed a dynamic oversampling approach with adaptive boosting (DysBoost) to address the CI problem in the quality control of a diesel engine manufacturing process.

Zhang et al. [126] (see Article 58 in Table 7) proposed, for machinery fault diagnosis, a novel data-driven method based on sample reliability assessment (SRA), and an improved convolutional neural network (ICNN). They tested them on the CRWU bearing fault dataset (see Section 5.5.1).

Pahren et al. [89] (see Article 59 in Table 7) proposed a class ordered synthetic minority oversampling technique (COSMOTE) to augment the class imbalanced data in quality control.

Yang et al. [249] (see Article 60 in Table 7) proposed a hybrid loss function of binary cross-entropy (BCE) [247] and Dice [248] loss functions in order to address the CI issue in welding defect detection with deep learning (DL-BCE-Dice). They use the GDXray dataset (see Section 5.5.3) for model test and evaluation, in particular, the X-ray welding image set with welding defects annotations.

Usuga-Cadavid et al. [240] (see Article 61 in Table 7) proposed to use a language model called GPT-2 [259] to generate artificial maintenance reports. They train a deep learning (DL) architecture named CamemBERT [239] used for harnessing data from historical maintenance logs. The CI problem is mitigated thanks to the artificial samples generated with GPT-2.

Li et al. [237] (see Article 62 in Table 7) proposed Balanced-YOLOv3, a CI training set/validation set split method deep learning algorithm for CI in object detection for the printed circuit board assembly. Their work is based on YOLOv3 [300]. Their strategy with Balanced-YOLOv3 includes an additional feature fusion for multi-scale feature imbalance, other than an efficient anchor concept to handle the CI.

Kuang et al. [242] (see Article 63 in Table 7) proposed a class-imbalance adversarial transfer learning (CIATL) deep learning network in order to learn domain-invariant knowledge in presence of imbalanced class data.

Dasari et al. [88] (see Article 64 in Table 7) proposed a cluster-based adaptive data augmentation (CADA), an oversampling algorithm to address the CI problem in additive manufacturing quality monitoring.

Hou et al. [114] (see Article 65 in Table 7) proposed, for the mechanical fault detection, a contrastive-weighted self-supervised model (CSM), together with an augmented vision transformer (VIT) [275], merging the imbalanced learning strategies at pre-training time. They compared their proposed CSM-VIT deep learning algorithm with other traditional data augmentation methods such as random oversampling (ROS) [3], random undersampling (RUS) [14] and re-weighting (RW) [117], finding out that long-tailed data (multi-minority-classes) classification is not compatible with the traditional methods. They also compared their proposed model with a self-supervised convolutional network (SSCN) [268], a residual network (ResNet) [265], a momentum contrastive network (MoCo) [263] and the advanced contrastive learning method of SimCLR [266].

Cho et al. [80] (see Article 66 in Table 7) compared a generative adversarial network (GAN) [28,29] (see Section 5.3.2) for the imbalanced data pre-processing with other five data-based algorithms, namely, random oversampling (ROS) [3], SMOTE [16], SMOTE with Tomek links (also known as SMOTE-Tomek [125]), SMOTE with edited nearest neighbor (ENN) [93] (also known as SMOTE-ENN) and adaptive synthetic sampling (ADASYN) [78] (see Section 5.1.1). The GAN produced the highest F1 score (see Section 5.4.7) on the SECOM dataset (see Section 5.5.5).

Song et al. [256] (see Article 67 in Table 7) proposed a latent mapping adversarial network (LMAN). They took inspiration from the style-based generative adversarial network (StyleGAN) [301] structure. They tested LMAN on the Severstal steel manufacturing process dataset (see Section 5.5.6) and assessed its better performance against three variants of generative adversarial networks (GAN) [28,29].

Kim et al. [251] (see Article 68 in Table 7) tested some ensembles of deep neural networks (DNN) [250,252] (DNN-Ens) for defect detection in the semiconductor manufacturing process, however, no comparisons were made.

Jalayer et al. [244] (see Article 69 in Table 7) used the Wasserstein generative adversarial with gradient penalty network (WGAN-GP) [278] in order to generate more of the rare training samples. They compared it with a weighted extreme learning machine (WELM) [276], and a convolutional long short-term memory (CLSTM) [243]. They tested their approach on the CRWU bearing fault dataset (see Section 5.5.1).

Sun et al. [205] (see Article 70 in Table 7) proposed to use the multiclass Mahalanobis–Taguchi system (MMTS) [204] for the fault diagnosis of industrial robots. The categories were classified with the deviation degree between the sample and the reference space. This approach appeared to be good for classifying CI data. They tested it on a private robot bearing fault dataset.

Nuhu et al. [81] (see Article 71 in Table 7) proposed a Borderline SMOTE SVM (BSMOTE-SVM) and compared it with other two data manipulation methods, namely, SMOTE and ADASYN. They tested the three DM techniques in combination with ten ML algorithms, namely,

Table 8
Most used data manipulation algorithms for class imbalance.

Abbr.	Algorithm	N. of articles
ADASYN	Adaptive synthetic sampling [78]	4 (5%)
IRS	Image resampling (we defined it in Section 5.1.2)	3 (4%)
ROS	Random oversampling [3]	5 (7%)
RUS	Random undersampling [14]	3 (4%)
SMOTE	Synthetic minority oversampling technique [16]	14 (19%)
SMOTE-ENN	SMOTE with edited nearest neighbor [124]	2 (3%)
US	Undersampling [3,130]	2 (3%)

AdaBoost DT, GBM, LDA, LR, MLP, NB, RF, SVM and XGBoost. They used the SECOM dataset (see Section 5.5.5).

Wu et al. [185] (see Article 72 in Table 7) proposed an inverse distance weights k-nearest neighbor (IDW-KNN) algorithm for the quality classification of powder splatter in additive manufacturing. The algorithm was presented in form of pseudo-code.

Jang and Kim [129] (see Article 73 in Table 7) proposed an unstructured borderline self-organizing map (UB-SOM) algorithm. They took advantage of the distributional information in borderline areas, and the key elements were feature selection and undersampling. They tested their solution on ten scientific datasets and two industrial datasets. The first industrial dataset is generated by simulating a dry etcher LAM TCP 9600 as done by Lee and Kim [302] and referred as etching dataset in their work. The second industrial dataset is SECOM (see Section 5.5.5).

Shen and Lee [236] (see Article 74 in Table 7) looked for the best data augmentation method to solve any CI problems between the defect classes of the semiconductor assembly process. They compared three data augmentation techniques with DL, namely the convolutional autoencoder (CAE) [238], the conditional variational autoencoder (CVAE) [246] and the auxiliary classifier generative adversarial network (ACGAN) [235].

5. Summary of results

5.1. Most used data manipulation algorithms

In Table 8, we report the seven DM algorithms, among the 37 found, that are most applied in manufacturing and we summarize them in the following subsections. The criterion for this selection is that the algorithm is applied at least in two of the reviewed articles.

5.1.1. Adaptive synthetic sampling

Adaptive synthetic sampling (ADASYN) [78] is an extension of SMOTE, an oversampling technique. It aims to balance the class distribution in imbalanced datasets by generating synthetic samples of the minority class. In regard to that, ADASYN uses a density-based approach to oversampling, which means that it generates more synthetic samples around the hard-to-classify samples, which are closer to the decision boundary and therefore more likely to be misclassified. This helps to improve the classification performance of the model on imbalanced datasets by increasing the diversity of the minority class samples and reducing the overfitting to the majority class.

5.1.2. Image resampling

Image resampling (IRS) is a form of data manipulation applicable to pre-processing of image data. To the best of our knowledge, there is no scientific article that classifies all the possible methods, thus we listed the major findings from the articles reviewed. In particular, Saqlain et al. [96] (see Article 8 in Table 6) performed random rotation of 10-degree, horizontal flipping, width shift, height shift, shearing range, channel shifting, and zooming on the input images; Park et al. [97] (see Article 11 in Table 6) randomly transformed and distorted raw image samples before feeding them to a CNN; Liang et al. [98] (see Article 23 in Table 6) applied region translation, random patch cropping, small angle image rotation, horizontal flipping, random contrast change, random brightness change and image resizing. The goal in IRS is to augment the number of images containing variations of the important features. The advantage is that images contain a good amount of information that can be easily manipulated to generate new perspectives of the same features.

5.1.3. Random oversampling

Random oversampling (ROS) [3] is a technique used in imbalanced data classification problems, where a minority class has fewer instances compared to the majority class. The method increases the number of samples in the minority class by randomly replicating instances from it. This approach is simple to implement but can lead to overfitting, as the same instances are repeatedly shown to the model.

5.1.4. Random undersampling

Random undersampling (RUS) [14] is a straightforward strategy focused on equalizing class distribution by randomly discarding instances from the majority class. Specifically, entries from the predominant class are indiscriminately pruned from the training dataset. While its simplicity and efficacy are commendable, a drawback is its nonchalant removal of samples, disregarding their potential significance in shaping the decision boundary between classes. A natural enhancement to RUS would be to exercise caution, ensuring pivotal majority class observations are not omitted, as their absence could compromise classification results.

5.1.5. Synthetic minority oversampling technique

The Synthetic Minority Oversampling Technique (SMOTE) [16] stands out as a widely adopted DM method, as highlighted in Table 8. At its core, SMOTE employs the k-nearest neighbors algorithm (KNN) to forge new instances for the minority class, thereby harmonizing the scales of majority and minority classes. The mechanism involves creating a synthetic data point by choosing a random midpoint between a given observation and one of its k-nearest neighbors. This method is iteratively applied, producing multiple synthetic data points and thus ensuring a dataset that boasts an equal representation of each class.

One of SMOTE's primary strengths is its ability to generate synthetic observations that echo the characteristics of existing minority class data, thereby retaining the dataset's inherent structure. This can pave the way for models that are not only more precise but also resilient, as these synthetic points serve as added training material for the model's learning process. However, it is crucial to recognize that while SMOTE effectively equilibrates the dataset, it does not directly tackle the root cause of CI. Hence, it is typically used synergistically with other ML or DL approaches to maximize performance outcomes.

5.1.6. Synthetic minority oversampling technique with edited nearest neighbor

Synthetic minority oversampling technique with edited nearest neighbor (SMOTE-ENN) [124] is a variation of the SMOTE method. It aims to improve the SMOTE method by addressing its tendency to generate synthetic samples that are too similar to existing minority class samples. This is done by applying the Edited Nearest Neighbor (ENN) method to the synthetic samples generated by SMOTE. The ENN method is used to remove those synthetic samples that are too similar to the majority class samples. This can help to improve the classification performance of models trained on the oversampled dataset. Despite its practical improvements to SMOTE, SMOTE-ENN seems to be still quite neglected in manufacturing (see Table 8).

Table 9
Most used machine learning algorithms for class imbalance.

Abbr.	Algorithm	N. of articles
1-SVM	One class support vector machine [134,135]	3 (4%)
AdaBoost	AdaBoost [139]	7 (9%)
Bagging	Bootstrap aggregating [146]	3 (4%)
BPN	Back-propagation neural network [150]	4 (5%)
DT	Decision tree [170]	3 (4%)
KNN	K-nearest neighbor (classifier) [188,189]	3 (4%)
LDA	Linear discriminant analysis [190,191]	3 (4%)
LR	Logistic regression [197]	3 (4%)
MLP	Multi-layer perceptron [202]	4 (5%)
PCA	Principal component analysis [215]	3 (4%)
RF	Random forest [221]	9 (12%)
RUSBoost	Random undersampling boost [23]	3 (4%)
SVM	Support vector machine [227]	6 (8%)

5.1.7. Undersampling

Undersampling (US) [3,130] is a technique employed to equilibrate datasets by diminishing the instances of the majority class to align with those of the minority class. Essentially, US entails the random omission of certain majority class instances, ensuring that both classes have equivalent representation in the resultant dataset. Yet, it is pivotal to acknowledge that US might inadvertently lead to information attrition, given the potential removal of valuable majority class instances. Thus, when leveraging US, a careful assessment balancing the advantages in performance against the potential loss of data becomes imperative.

5.2. Most used machine learning algorithms

In Table 9, we report 13 ML algorithms, among the 89 found, that are most applied in manufacturing and we summarize them in the following subsections. The criterion for this selection is that the algorithm is applied at least in three of the reviewed articles.

5.2.1. One class support vector machine

The One-Class Support Vector Machine (1-SVM) [134,135] is a specialized variant of SVM tailored for detecting anomalies. At its heart, the 1-SVM seeks to delineate a boundary that segregates the bulk of data points from the few outlier instances. During its training phase, the objective is to carve out a boundary that maximizes the distance (or margin) between itself and the nearest points of the predominant class. This distinctive boundary is influenced by a select group of data points, termed as support vectors, which lie closest to it.

In the prediction stage, an incoming data point's affiliation – whether it belongs to the majority cluster or is an anomaly – is determined by its position relative to this boundary. Data points residing on one side are perceived as part of the main cohort, while those on the opposite end are flagged as anomalies. The real strength of 1-SVM emerges in scenarios where the primary class is distinctly characterized, but the anomalous (or minority) class remains elusive, making it apt for tackling imbalanced datasets.

5.2.2. AdaBoost

AdaBoost [139] is a type of ensemble learning technique. It operates iteratively, capitalizing on the errors of weak classifiers – those that achieve just a tad better than mere random predictions – and amplifies their capabilities to mold them into potent classifiers. The essence of AdaBoost lies in its ability to amalgamate several weak classifiers, enhancing the ones that showcase superior performance through higher weight assignments. Although AdaBoost was not crafted explicitly for skewed datasets, its knack for allocating increased weights to under-represented classes allows it to show promise in addressing CI. This ensures the algorithm zeroes in on accurately categorizing these lesser-represented samples. However, highly skewed class distributions can still present hurdles for the method.

5.2.3. Bootstrap aggregating

Bootstrap aggregating, commonly known as Bagging [146], is a strategy aimed at enhancing a machine learning model's performance. It achieves this by training several iterations of the model on distinct data subsets, subsequently amalgamating their predictions. For regression tasks, these predictions are averaged, while for classification, majority voting is employed.

The underlying philosophy of bagging is rooted in its approach: by instructing numerous models on varied data fragments, each model inherently deviates slightly from the others, exhibiting distinct biases and variances. By pooling or averaging the outputs of these diverse models, the combined model's overall performance is heightened, with both bias and variance experiencing a reduction. In scenarios with imbalanced datasets, bagging can be particularly potent if data subsets are curated to maintain a balanced representation between the minority and majority classes.

5.2.4. Backpropagation Neural Network

A Backpropagation Neural Network (BPN) [150] is a type of Artificial Neural Network (ANN) that undergoes training through a supervised learning technique known as backpropagation, as further detailed in Section 5.2.9 concerning the Multi-Layer Perceptron. In scenarios with CI, the backpropagation method can be synergized with strategies like weight balancing or sample re-sampling. Such integrations amplify the significance of the minority class during the training phase. This strategic emphasis ensures that the network does not lean excessively towards the majority class, subsequently enhancing its overall precision.

5.2.5. Decision Tree

A Decision Tree (DT) [170] serves as a tool for decision-making and classification. Structurally resembling a tree, it comprises a succession of choices and their subsequent outcomes. Every node in the tree signifies a test on a particular attribute, each branch depicts the test's outcome, and each terminal or leaf node denotes a class label or a decision. Versatile in nature, DTs can process both categorical and numerical data, offering the benefits of easy interpretation and visualization. The tree's formation hinges on a recursive partitioning technique, wherein the algorithm pinpoints the attribute that optimally divides the data into uniform subsets. This partitioning persists until certain criteria, such as a predefined number of samples in a subset or a set tree depth, are met.

Owing to their recursive data-splitting approach based on varying attributes, DTs excel at uncovering decision boundaries adept at distinguishing between minority and majority classes. Being non-parametric, they do not predicate on any presumptions about the data's inherent distribution. This grants them the flexibility to capture intricate relationships and patterns, making them potentially apt for datasets where the minority class showcases distinct traits.

However, DTs are not without limitations. They can display a bias towards the dominant class, especially when imbalances are stark. Additionally, their tendency to not generalize optimally and susceptibility to overfitting can be potential pitfalls.

5.2.6. K-nearest neighbor

The K-nearest neighbor (KNN) algorithm [188,189] operates within supervised learning, serving to assign a label to an unidentified data point using pre-labeled data. KNN's foundational principle is to categorize a specific test data point based on the majority label amongst its "k" closest neighbors in the training dataset. The value of "k" is user-determined and is usually a minimal figure, like 1 or 3. For every incoming sample, KNN pinpoints the k closest training samples within the feature space, generally gauging distances using the Euclidean metric. The algorithm then bestows upon the new sample the predominant class label among its k proximal training counterparts.

KNN's versatility allows it to cater to both classification and regression tasks. While classification tasks yield a class label as an outcome, regression produces a continuous value. In both scenarios, the objective is to identify the most analogous observations from the training set for an incoming data point. The prediction leans on the most recurrent output (be it class label or continuous value) among the neighboring training samples.

KNN proves particularly advantageous with imbalanced datasets. It offers a mechanism to allocate the likeliest class to a freshly introduced sample. The utility of this process emerges when, for instance, new minority class samples are incorporated, while new majority class samples are omitted. Such an approach is harnessed and refined in DM methods tailored for data augmentation, like SMOTE or SMOTE-ENN.

5.2.7. Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) [190,191] is a technique aimed at reducing dimensionality. It strives to project data onto a lower-dimensional plane, ensuring the distinctions between classes remain intact. Tailored for classification tasks, LDA's objective is to unearth a linear blend of features that adeptly differentiates between classes. The methodology involves augmenting the ratio of variance between classes to the variance within classes, a principle known as the Fisher criterion.

LDA proves invaluable when navigating data with copious dimensions. It aids in truncating the feature count while safeguarding the vital classification information. A couple of foundational assumptions underpinning LDA include uniform covariance matrices across classes and normally distributed features. If the data diverges from these assumptions, LDA might not deliver optimal outcomes.

However, one of LDA's strengths is its resilience to CI, especially if class means and covariances are pinpointed with precision. Its operation pivots on the covariance matrices inherent to the classes and eschews making direct presumptions about class distributions. As a result, LDA remains adept at extracting distinguishing details from both minority and majority classes, even in the face of dataset imbalances.

5.2.8. Logistic Regression

Logistic Regression (LR) [197] is a statistical approach crafted to ascertain the likelihood of a binary outcome based on one or more predictors. It accomplishes this by pinpointing the optimal coefficients for these predictors, which escalates the probability of the binary outcome being categorized as 1. Subsequently, these coefficients facilitate predictions for novel data points. One of LR's merits is its ability to yield probability estimates for each class, which can be construed as confidence levels. This flexibility in setting decision boundaries proves advantageous in skewed datasets, especially when the minority class demands attention, facilitating the recalibration of the decision threshold to favor sensitivity over specificity.

However, LR is not devoid of challenges. It manifests an inclination towards the dominant class, potentially compromising its efficacy in accurately identifying minority class entities. Moreover, it predicates on a linear decision frontier, which might falter in deciphering intricate relationships in imbalanced datasets, especially if the minority class showcases non-linear dynamics. Additionally, LR operates on the presumption of a linear correlation between features and the log-odds of the resultant variable. A pronounced disparity in feature distribution between majority and minority classes could impede LR's performance.

To circumvent these pitfalls, LR is optimally employed in tandem with other strategies. Adjusting class weights can counterbalance the bias slanting towards the predominant class. Resampling techniques like SMOTE or undersampling methods like RUS, ensemble techniques, and tweaking thresholds can render the model more receptive to the minority class. This, however, might raise the specter of heightened false positives.

5.2.9. Multi-layer Perceptron

The Multi-layer Perceptron (MLP) [202] is a specific variant of the feedforward (FF) Artificial Neural Network (ANN). It boasts a structure comprising multiple layers of synthetic neurons, known as "perceptrons", interconnected and sequentially arranged. The input layer assimilates the primary data, and the subsequent layers progressively manipulate and transfigure this data via a cascade of computations. The terminal layer, termed the output layer, furnishes the model's prognostications. Connection weights spanning the layers undergo refinement during the training phase, facilitated by a technique called backpropagation (further elaborated in Section 5.2.4 pertaining to the Backpropagation Neural Network). This method tweaks the weights to diminish the discrepancy between the model's predictions and the genuine target outcomes.

MLPs excel at deciphering intricate non-linear interrelations between features and class identifiers. Such prowess proves invaluable for skewed datasets, which might manifest complex configurations. While MLPs in their native form are not tailored for CI, their amalgamation with ensemble techniques like bagging or boosting augments their efficacy in handling imbalanced datasets. Such ensemble configurations bolster generalization and adeptly address CI, either by tutoring multiple MLPs on diverse data fragments or by recalibrating the weights of erroneously classified entities.

5.2.10. Principal Component Analysis

Principal Component Analysis (PCA) [215] stands out as a prevalent method for dimensionality reduction, adeptly converting data from a high-dimensional space to a more manageable lower-dimensional format. Often employed as an initial step for visual representation and subsequent analyses, PCA does not inherently cater to CI. However, when synergized with other strategies, such as oversampling or undersampling, it becomes conducive to addressing the challenges posed by imbalanced datasets.

5.2.11. Random Forest

Random Forest (RF) [221] is an ensemble-based learning method tailored for classification and regression tasks. It synergizes multiple decision trees, crafting a model that is both robust and precise. Within RF, each tree undergoes training on a distinct subset of data, and at every tree node, a random feature subset serves as the split determinant. The ultimate prediction materializes either by averaging the tree outputs (in regression) or through a majority consensus (in classification). Such randomization fortifies the model against overfitting and heightens its stability.

RFs are championed for their exceptional accuracy, adaptability, and capability to navigate extensive datasets riddled with missing entries and erratic features. They are equipped to manage both binary and multi-class classification scenarios, as well as regression challenges. One of RF's strengths is its inherent resilience to CI. By weaving together numerous decision trees and amalgamating their verdicts, RF can diminish the repercussions of skewed class distributions and curtail any bias leaning towards the dominant class. RF's architecture also enables it to spot anomalies within the dataset by scrutinizing individual tree behaviors, an attribute that is especially beneficial when the minority class encapsulates rare or atypical samples.

However, while RF is adept at contending with CI, it is also resource-intensive. Furthermore, its potential inclination towards the majority class, given its predominance in the dataset, might skew the model's sensitivity and specificity. Such biases could culminate in challenges when striving to accurately pinpoint the minority class.

Table 10
Most used deep learning algorithms for class imbalance.

Abbr.	Algorithm	N. of articles
CNN	Convolutional neural network [245]	9 (12%)
GAN	Generative adversarial network [28,29]	5 (7%)
ResNet	Residual network [265]	2 (3%)

5.2.12. Random Undersampling Boost

RUSBoost [23] is an innovative algorithm that fuses Random Undersampling (RUS) with the boosting ensemble technique to adeptly manage CI in datasets. Boosting operates by sequentially constructing weak models, which are later amalgamated to give rise to a robust model. Within the framework of RUSBoost, the initial step involves undersampling the training dataset to equilibrate class distributions. Subsequently, using the boosting approach, weak models are successively developed on this balanced dataset. Each subsequent model zeroes in on samples that previous models misclassified. The culmination of this process yields a potent model derived from the collective results of all individual weak models.

RUSBoost has exhibited proficiency in addressing CI quandaries, particularly in scenarios where the minority class is scanty and the majority class is voluminous.

5.2.13. Support Vector Machine

Support Vector Machine (SVM) [227] is a supervised learning technique tailored for both classification and regression challenges. In its essence, SVM identifies a hyperplane – a linear demarcation – in a multidimensional space to optimally segregate data into distinct classes. Notably, the data points in closest proximity to this hyperplane are termed 'support vectors' and play a pivotal role in delineating the class boundaries. Once this boundary is established, classifying new data becomes straightforward, contingent on which side of the hyperplane the data resides. For regression tasks, SVM strives to pinpoint a hyperplane that most congruently aligns with the data.

In scenarios riddled with CI, SVM proves invaluable. It excels at discerning the spatial region inhabited by the minority class, facilitating the synthesis of new samples essential for dataset equilibration. SVM seamlessly collaborates with other Data Manipulation (DM) techniques in this endeavor. Additionally, its binary incarnation, the one-class SVM (1-SVM), serves as a potent outlier detection tool, enhancing the classification of minority class instances.

5.3. Most used deep learning algorithms

In Table 10, we report the three DL algorithms, among the 44 found, that are most applied in manufacturing and we summarize them in the following subsections. The criterion for this selection is that each selected algorithm is applied at least in two of the reviewed articles.

5.3.1. Convolutional Neural Network

Convolutional Neural Networks (CNN) [245] are specialized deep learning models primarily harnessed for image-oriented tasks like classification, object detection, and segmentation. Their architecture is intrinsically tailored for image data, capturing its spatial intricacies. A typical CNN is architecturally layered, encompassing convolutional layers, pooling layers, and fully connected layers. The convolutional strata apply filters to incoming images, pinpointing specific features like edges and textures. Pooling layers, meanwhile, compact the feature maps' spatial dimensions, retaining vital information. The fully connected layers interpret these gleaned features to arrive at a definitive prediction.

Training CNNs involves leveraging vast quantities of labeled data in tandem with supervised algorithms, notably backpropagation. Their prowess has been manifest in an array of computer vision benchmarks, cementing their utility in diverse sectors, including wafer and semiconductor manufacturing.

CNNs are adept at hierarchically processing data. This proficiency is invaluable for skewed datasets, as CNNs autonomously discern vital features across various abstraction layers, encapsulating crucial patterns and diminishing the ramifications of CI. By employing local receptive fields and uniform weight distribution, CNNs discern spatial and temporal data patterns. This setup permits them to recognize local dependencies and capitalize on translational and rotational consistencies, making them resilient to object variations. Such resilience is instrumental for imbalanced datasets teeming with varied minority class instances. Especially when imbalanced datasets undergo augmentation via Data Manipulation (DM) techniques like IRS (elaborated further in Section 5.1.2), CNNs prove beneficial.

Furthermore, leveraging pre-trained CNN models, such as those nurtured on expansive image databases like ImageNet, offers a foundational advantage for imbalanced datasets. Through transfer learning, the vast knowledge from these expansive datasets can be channeled to enhance the performance on imbalanced datasets that may be data-scarce.

5.3.2. Generative Adversarial Networks

Generative Adversarial Networks (GAN) [28,29] comprise a sophisticated deep learning architecture with a dual-component structure: a generator and a discriminator. The generator's role revolves around crafting data samples reminiscent of genuine training data, whereas the discriminator's mission is to discern between these generated samples and the authentic training specimens.

In its operational flow, the generator consumes random noise to produce novel data samples. These synthesized samples, alongside the genuine training data, are subsequently fed to the discriminator, which allocates a probability score reflecting the likelihood of each sample being genuine. Both components – generator and discriminator – are co-trained. The generator aspires to fabricate samples that the discriminator perceives as authentic, while the discriminator endeavors to accurately segregate real from synthesized samples. This iterative training dynamic persists until the generator's output becomes virtually indistinguishable from the original training set. Once this equilibrium is attained, the generator can conjure new, congruent data samples that were not part of the initial training set.

GANs have found applications spanning diverse domains, from image synthesis to text-to-speech conversions. They stand out particularly in generating new minority class samples to address dataset imbalances. Effectively, GANs epitomize the deep learning progression of Data Manipulation (DM) techniques, offering a dedicated deep learning framework exclusively crafted for synthetic sample generation.

5.3.3. Residual Network

The Residual Network (ResNet) [265] emerges as a specialized variant of deep convolutional neural networks (CNN). Its inception aimed to tackle the vanishing gradient conundrum, a prevalent issue in extremely deep networks. This problem manifests as dwindling gradient values, complicating the network's training process. ResNet introduces an innovative solution via residual connections that sidestep one or more layers. These connections facilitate the network in discerning the residual relationship between inputs and outputs, rather than comprehending the complete mapping. This enhancement not only simplifies learning but also expedites the training regimen.

Owing to these residual connections, ResNets can delve into far deeper architectures, boasting over 100 layers, surpassing traditional CNNs. Demonstrating superior performance on various computer vision benchmarks, ResNets have become a staple for tasks like image classification, object detection, and semantic segmentation. Their prowess extends beyond vision, finding applications in natural language processing, among others.

Table 11
Confusion matrix.

	Actual positive	Actual negative
Predicted positive	TP	FP
Predicted negative	FN	TN

Building on their versatility, ResNets can also exploit pre-trained models, particularly those honed on expansive image datasets such as ImageNet. By harnessing transfer learning, these pre-trained ResNet models can imbue a foundation of rich features into imbalanced datasets, promoting better generalization and enhanced performance, even when training data is sparse.

5.4. Most used metrics for class imbalance solutions

In the examined papers, we found several metrics adopted to evaluate the quality of the algorithms dealing with CI data. Each metric is presented in a separate subsection below, and the most used ones are reported in Table 12.

5.4.1. Confusion matrix

The confusion matrix (see Table 11) helps us to define four parameters that are the base for calculating other metrics, namely, true positives (TP), false positives (FP), false negatives (FN) and true negatives (TN). We can generate a confusion matrix at the end of a classification task by comparing all the classified samples with their true classes (which we already knew). For this purpose, the test set provides the necessary data.

5.4.2. Accuracy

Accuracy is a metric for classification models that measures the number of predictions that are correct as a percentage of the total number of predictions that are made.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

5.4.3. Sensitivity

Sensitivity in machine learning gauges the model's proficiency in correctly identifying positive cases. Often referred to as the true positive rate **(TPR)** or **recall**, this metric highlights the model's ability to recognize positive samples accurately.

$$Sensitivity = \frac{TP}{TP + FN}$$

5.4.4. Specificity

Specificity, in the context of machine learning, refers to the model's ability to correctly identify negative instances. Often termed the true negative rate **(TNR)**, this metric underscores the model's precision in distinguishing negative samples from positive ones.

$$Specificity = \frac{TN}{TN + FP}$$

5.4.5. Precision

Precision emphasizes the accurate identification of the positive class, gauging the likelihood of correctly pinpointing positive values. Specifically, it calculates the fraction of true positives from those labeled as positive by the model. Notably, precision remains unaffected by an overwhelming presence of negative samples, highlighting its relevance in situations with CI.

$$Precision = \frac{TP}{TP + FP}$$

5.4.6. Geometric mean

The geometric mean (G-Mean) evaluates the equilibrium in classification outcomes for both predominant and less frequent classes. When G-Mean is low, it signals that the model may struggle with correctly identifying positive instances, even if it excels at classifying the negative ones. Employing G-Mean is crucial to circumvent the pitfalls of excessively tailoring to the negative class while neglecting the positive class.

$$Geometric\ mean = \sqrt{Precision * Recall}$$

5.4.7. F1 score

The F1 score is defined as the harmonic mean of precision and recall, because it combines the precision and recall metrics into a single metric, giving equal weight to precision and recall.

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall}$$

5.4.8. Area under the ROC curve

The Receiver Operating Characteristic (ROC) is a prevalent graph employed to assess binary classifier systems. It depicts the relationship between the true positive rate and the false positive rate across different threshold levels, facilitating a visual understanding of their interplay. Through the ROC curve, one can pinpoint the best threshold for a given system. The area beneath this curve, termed the Area Under the Curve (AUC), serves as a consolidated indicator of the classifier's efficacy.

5.4.9. FPR

The False Positive Rate (FPR) calculates the proportion of negative instances that a classifier mistakenly labels as positive within a binary classification scenario. Essentially, it measures the frequency with which a classifier misidentifies negative samples. FPR serves as a key metric for gauging a classifier's performance and typically occupies the x-axis in a ROC curve representation. A reduced FPR suggests heightened specificity, denoting the classifier's proficiency in accurately recognizing negative cases.

$$FPR = \frac{FP}{FP + TN}$$

where FP are the number of cases that are incorrectly classified as positive, and TN are the number of cases that are correctly classified as negative.

5.4.10. FNR

The False Negative Rate (FNR) quantifies the proportion of positive cases that a classifier erroneously labels as negative in a binary classification context. Essentially, it gauges the frequency at which a classifier misses positive instances. FNR is a crucial metric for assessing a classifier's performance and stands as the inverse of the True Positive Rate (TPR). A diminished FNR suggests heightened sensitivity, signifying adeptness in accurately detecting positive cases.

$$FNR = \frac{FN}{TP + FN}$$

where FN are the number of cases that are incorrectly classified as negative, and TP are the number of cases that are correctly classified as positive.

5.5. Publicly available class imbalanced datasets for benchmarking in manufacturing

This section contains a list of the eleven publicly available CI datasets found by the systematic review of the selected articles (see Table 5). It is important, for benchmarking purposes, that the data used to evaluate the proposed algorithms is also available to reproduce past results in future research. The most used dataset is SECOM (see Section 5.5.5), found in seven articles about semiconductor manufacturing. The second most used is WM811K (see Section 5.5.11), used in

Table 12
Most used metrics for class imbalance.

Abbr.	Metric	N. of articles
–	Accuracy	49 (66%)
–	Sensitivity	37 (50%)
F1	F1 score	25 (34%)
AUC	Area under the ROC curve	24 (23%)
–	Specificity	19 (26%)
G-Mean	Geometric mean	17 (23%)
–	Precision	16 (22%)
FPR	False positive rate	10 (14%)
FNR	False negative rate	7 (9%)

Table 13
Most used public datasets for class imbalance.

Abbr.	Dataset	N. of articles
SECOM	Semiconductor manufacturing process [283]	7 (9%)
WM811K	Wafer maps [289]	5 (7%)
CRWU bearing fault	Case Western Reserve University bearing fault [279]	4 (5%)
SPF	Steel plates faults from UCI machine learning repository [285]	2 (3%)

five articles about wafer maps, also in semiconductor manufacturing. Although we report in Table 13 the four most used public datasets that we found in the articles, with a criterion of at least two articles citing the same dataset, we provide below, in alphabetic order, a short description of each of all the eleven public datasets that we found, because we believe they have a high value for future benchmarkings.

5.5.1. CRWU bearing fault dataset

The Case Western Reserve University (CRWU) bearing fault dataset [279] contains signals under four different bearing states, including normal, inner race fault, outer race fault, and rolling element fault. This dataset is used by Zhang et al. [126], Jalayer et al. [244], Wei et al. [77] and Pan et al. [253].

5.5.2. GC10-DET dataset

GC10-DET [280] is a dataset that contains ten manufacturing defect types, including punching hole, crease, crescent gap, silk spot, rolled pit, inclusion, oil spot, welding line, water spot, and waist folding. It has limited samples in minority class Cr and RP. This dataset is used by Wang et al. [234].

5.5.3. GDXray dataset

The GDXray dataset [281] includes five image groups: castings, welds, baggage, nature objects, and settings. This dataset is used by Yang et al. [249].

5.5.4. NEU-CLS-64 dataset

NEU-CLS-64 [282] is a dataset with nine manufacturing defects that are obtained by manual selection and cropping, including pitted surface, crazing, inclusion, patches, grooves and gouges, rolling dust, rolled-in scale, scratches and spots. This dataset is used by Wang et al. [234].

5.5.5. Semiconductor manufacturing process dataset

The semiconductor manufacturing process dataset [283], also known as SECOM, contains 1567 instances of 594 features, with 1463 *pass* and 104 *fail* samples. The CI ratio is 1:14. This dataset is used by Wang et al. [187], Nuhu et al. [81], Chazhoor et al. [123], Kim et al. [157], Wang et al. [133], Wang et al. [187], Cho et al. [80] and Jang and Kim [129].

5.5.6. Severstal steel manufacturing process dataset

The Severstal steel manufacturing process dataset [284] is a collection during the production process of a high-frequency images of flat sheet steel. The dataset contains multiple classes of defect-type data, a single class of defect-type data, and non-defect-type data, leading to different ratios. The defect data consists of 5680 samples that correspond to 53.04% of the data. The proportions of each class are as follows: 72.59% (4123 samples) for pitted surface and scratch, 12.64% (718 samples) for crazing, 11.29% (641 samples) for inclusion, and 3.48% (198 samples) for rolled-in scale. This dataset is used by Song et al. [256].

5.5.7. SPF dataset

The Steel Plates Faults (SPF) dataset of the UCI Machine Learning Repository [285] contains 1,941 instances with 27 features and 7 types of steel plates faults. This dataset is used by Lee et al. [255], and Lee and Baek [254].

5.5.8. SW dataset

The semiconductor wafer (SW) dataset is provided by the UEA & UCR Time Series Classification Repository [286]. The SW dataset contains 152 sensor values taken from a semiconductor manufacturing wafer process. The CI ratio is of 762 abnormal samples (10.7%) out of 7,164 total samples. This dataset is used by Song and Baek [113] who also present an exploit for enhancing CI: in order to match the 3-sigma of the actual manufacturing process, they conduct their experiments using only 30 abnormal samples from the SW dataset.

5.5.9. TD dataset

The textile defects (TD) dataset [287] contains ten classes of textile defect types. This dataset is used by Wang et al. [234].

5.5.10. Wood dataset

The wood dataset [288] contains seven wood defect types, including dry knot, encased knot, horn knot, decayed knot, leaf knot, edge knot, sound knot. This dataset is used by Wang et al. [234].

5.5.11. WM811K dataset

The WM811K dataset [289], known as the largest known wafer map dataset available to the public, contains 811,457 wafer maps provided by Taiwan Semiconductor Manufacturing Company and specifies for each the failure type. However, the CI ratio is unspecified. This dataset is used by Saqlain et al. [96], di Bella et al. [109], Yu et al. [152] and Yu and Liu [262].

5.6. Publicly available source code for CI solutions in manufacturing

We reviewed 74 articles addressing CI problems in the manufacturing domain to investigate the availability of repositories and publicly accessible source code. Surprisingly, none of the reviewed articles included hyperlinks or references to any repositories or to any publicly available source code, indicating a significant gap in code availability. This finding raises concerns regarding the replicability, transparency, and reproducibility of the proposed algorithms.

However, there were some positive findings in terms of providing algorithmic insights. Out of the 74 articles, 21 of them included pseudo-code accompanying their proposed algorithms. Pseudo-code offers a high-level representation of the algorithmic steps employed in the studies, allowing readers to gain an understanding of the proposed methods. While pseudo-code does not provide the same level of detail as actual source code, it can still aid in replication and implementation to some extent.

6. Discussion

In this section we provide the major discussion points divided in subsections corresponding to our article's research questions. Our research produced an overview of 74 articles pertaining CI problems and their relative solutions in the manufacturing domain. Firstly, we explain what are the operations that are most affected by CI problems. Secondly, we provide our considerations about the most applied solutions to these CI problems. Thirdly, we discuss the most used metrics to verify the quality of the CI solutions. Lastly, we debate on the general (un)availability of public code and datasets for benchmarking purposes.

6.1. RQ1 on manufacturing operations affected by CI problems

Our review of CI problems in the manufacturing domain has yielded significant insights regarding the most prevalent manufacturing tasks where CI is observed. Notably, we identified several key areas where CI is particularly prominent, including defect detection, quality control, anomaly detection, and predictive maintenance, among others. Recognizing these specific areas of CI is crucial for researchers and practitioners, as it allows them to focus their efforts and resources on developing targeted and effective solutions tailored to address the challenges associated with these manufacturing tasks.

6.1.1. Defect detection

CI often arises in manufacturing processes when the occurrence of defective products is relatively low compared to the number of non-defective products. Detecting and accurately classifying these rare defective instances is of utmost importance to ensure product quality and minimize waste. By acknowledging the presence of CI in defect detection tasks, researchers can design algorithms and models that effectively handle the imbalance and enhance the accuracy and efficiency of defect identification systems.

6.1.2. Quality control

Quality control in manufacturing involves assessing the conformity of products to predefined standards or specifications. Imbalanced data can arise when the occurrence of non-conforming or faulty products is significantly lower than the occurrence of conforming products. Addressing CI in quality control tasks is crucial to avoid false positives or false negatives, ensuring accurate identification of non-conforming products and maintaining high-quality standards. By recognizing this challenge, researchers can develop specialized approaches that consider the CI to improve the overall performance of quality control systems.

6.1.3. Anomaly detection

Anomaly detection plays a vital role in manufacturing to identify deviations from normal or expected behavior in various processes or equipment. Imbalanced data often emerges when the occurrence of anomalies is relatively rare compared to normal instances. Accurate identification of anomalies is critical to prevent equipment failures, reduce downtime, and enhance overall operational efficiency. Recognizing CI in anomaly detection tasks enables researchers to develop robust models that can effectively handle the rarity of anomalies and improve the reliability and accuracy of anomaly detection systems.

6.1.4. Predictive maintenance

Predictive maintenance aims to anticipate equipment failures or malfunctions before they occur, thereby minimizing downtime and reducing maintenance costs. CI commonly arises in predictive maintenance tasks due to the low occurrence of failure events compared to normal operating conditions. By acknowledging the CI, researchers can focus on developing methods that effectively address this challenge, enabling accurate prediction and timely maintenance actions.

By identifying these specific manufacturing tasks where CI is prevalent, our review provides valuable guidance for researchers and practitioners in directing their efforts towards developing targeted solutions. Recognizing the unique characteristics and challenges posed by CI in defect detection, quality control, anomaly detection, and predictive maintenance allows for the development of more accurate, efficient, and reliable systems tailored to the specific needs of manufacturing processes. These findings pave the way for the advancement of effective solutions and contribute to improved manufacturing operations, quality assurance, and maintenance practices.

6.2. RQ2 on the solutions applied to CI problems in manufacturing

Our review of CI problems in manufacturing highlights the significance of data manipulation techniques in rebalancing the class distribution and enhancing the performance of classifiers. Imbalanced datasets, where the number of instances in different classes is heavily skewed, can hinder the effectiveness of machine learning models. DM techniques, such as SMOTE, ADASYN, ROS and RUS, aim to address this issue by adjusting the class distribution, ensuring sufficient representation of minority classes during the training process. We also found a new category of DM algorithms that we named image resampling or IRS, predominant in the semiconductor manufacturing with computer vision ML and DL algorithms. All these techniques play a crucial role in mitigating the impact of CI and improving the accuracy and reliability of classifiers in manufacturing applications.

In addition to DM techniques, ML and DL techniques have emerged as powerful tools. These approaches leverage algorithms that can effectively learn from imbalanced datasets and make accurate predictions. By employing techniques such as ensemble methods (e.g., RF and AdaBoost), cost-sensitive learning, or weighted loss functions, machine learning models can effectively handle CI and provide reliable predictions, even in scenarios where the minority class is underrepresented. Deep learning models, in particular CNNs, with their ability to automatically extract complex features, have shown promise in handling CI problems in manufacturing tasks. GANs also have a powerful use in generating synthetic samples to re-balance the CI datasets. The widespread adoption of these techniques signifies their effectiveness in addressing CI challenges and improving the overall performance of manufacturing systems.

To effectively tackle CI problems in manufacturing, it is essential for researchers to understand the underlying principles of DM, ML, and DL techniques. Familiarity with the concepts, strengths, and limitations of these approaches can guide researchers in selecting appropriate methods for their specific manufacturing contexts. Different tasks and scenarios may require tailored solutions, and the choice of technique should align with the specific objectives and constraints of the manufacturing application. Understanding the principles of these techniques empowers researchers to make informed decisions and optimize their approach to effectively address CI challenges.

Researchers should consider several practical factors when selecting appropriate methods to address CI in manufacturing. These factors include the size of the dataset, severity of CI, computational resources, interpretability requirements, and the complexity of the problem. For example, oversampling techniques, e.g., SMOTE, can be effective for increasing the representation of minority classes, but they may also introduce synthetic data that might affect model interpretability. Undersampling techniques can improve training efficiency but may discard potentially valuable information. Balancing these considerations is crucial to ensure that the chosen techniques align with the specific manufacturing context and achieve the desired balance between performance and practical constraints.

6.3. RQ3 on metrics for CI problems in manufacturing

In the context of our review article focusing on CI problems in manufacturing, it is paramount to address the limitations of accuracy as a metric when dealing with imbalanced datasets. Although accuracy is a commonly employed metric, it may prove to be the least accurate in scenarios where CI exists. Given the prevalence of CI issues in manufacturing datasets, it becomes imperative to explore alternative evaluation metrics that are better suited for assessing the performance of algorithms in such contexts.

One of the key considerations when evaluating algorithms in the manufacturing domain is the ability to detect and predict rare events or minority classes accurately. CI often leads to skewed datasets, where the occurrence of the minority class is significantly lower than that of the majority class. In such cases, accuracy can be misleading as a performance measure because a classifier that labels all instances as the majority class will still achieve a high accuracy due to the disproportionate number of majority class instances. This highlights the need to go beyond accuracy and delve into metrics that provide a more comprehensive understanding of model performance.

Metrics such as precision, recall, F1-score, and AUC offer valuable insights into the algorithm's performance in imbalanced manufacturing datasets. Precision measures the proportion of correctly predicted positive instances out of all instances predicted as positive, thus emphasizing the ability to identify the minority class accurately. Recall, on the other hand, focuses on the proportion of correctly predicted positive instances out of all actual positive instances, highlighting the algorithm's ability to capture the true positive cases. F1-score combines both precision and recall, providing a balanced measure that considers both metrics simultaneously.

Moreover, AUC offers a comprehensive evaluation of the algorithm's performance across various classification thresholds. This metric takes into account the trade-off between precision and recall and provides a holistic view of the classifier's discriminative ability, particularly in the presence of imbalanced classes. Studies in the manufacturing domain should include these alternative metrics to ensure a thorough evaluation of the proposed algorithms' efficacy.

By adopting appropriate evaluation metrics beyond accuracy, researchers can shed light on the performance of algorithms in tackling CI problems in manufacturing. This comprehensive analysis will enable better-informed decision-making, identification of successful approaches, and ultimately contribute to the advancement of effective solutions in the manufacturing sector.

6.4. RQ4 on public code and datasets for CI problems in manufacturing

One notable finding in our review of articles addressing CI problems in manufacturing is the scarcity of publicly available source code. Despite the growing interest in this area, we observed a dearth of articles providing access to the source code used in their research. This absence poses a significant challenge for the reproducibility and transparency of the proposed methods and algorithms. It hinders other researchers and practitioners from replicating and building upon the reported findings, potentially impeding progress in this field. Addressing this limitation and encouraging the open sharing of code would greatly enhance the rigor and advancement of research in CI problems in manufacturing.

Another key observation from our review is the limited availability of benchmarking datasets specifically tailored to CI problems in the manufacturing domain. While numerous articles address CI issues, we found that only a small subset provided publicly accessible datasets for evaluating and comparing different algorithms. This scarcity poses challenges in benchmarking the performance of various approaches and hampers the establishment of standardized evaluation procedures. To facilitate progress in this field, it is crucial to develop and share benchmarking datasets that accurately reflect the intricacies of CI in manufacturing. Such datasets would enable fair comparisons between different methods and contribute to the development of robust and effective solutions.

The absence of publicly available source code and limited benchmarking datasets has implications for the replicability and generalizability of research findings in the context of CI problems in manufacturing. Without access to the source code, reproducing the reported results and verifying the effectiveness of the proposed methods becomes challenging. Moreover, the limited availability of benchmarking datasets restricts the assessment of algorithm performance across a wide range of manufacturing scenarios. These limitations hinder the validation and wider applicability of research findings, which are critical for industrial adoption and real-world implementation.

The identified gaps in the availability of source code and benchmarking datasets present opportunities for collaboration and data sharing within the research community. Encouraging researchers to openly share their source code and datasets would foster a culture of collaboration, enable cross-validation of findings, and facilitate the advancement of CI solutions in the manufacturing domain. Initiatives promoting code repositories, data sharing platforms, and standardized evaluation procedures could help address these challenges and promote more robust research practices.

7. Future research directions

This section summarizes the future research directions outlined in this review article.

From our first research question, RQ1, it turns out that the manufacturing operations mostly affected by CI problems are defect detection, quality control, anomaly detection and predictive maintenance. Future works in these areas, but not limited to them, should necessarily acknowledge and address the presence of CI. Furthermore, we invite all the researchers and practitioners to study the ratio of positive and negative elements in their data (i.e., the CI ratio) before attempting any use of ML algorithms to classify, detect, predict or perform any learning operations in manufacturing. While data science can provide many solutions for different data configurations and use cases, this review proves how it is necessary for practitioners to be able to learn to apply these techniques with strong contextualization, and a bit of creativity, in the manufacturing domain. We hope that future works can use our article as a primary reference, and hopefully continue adapting other existing algorithms to ML operations in the manufacturing domain.

On a side note, the amount of DM, ML, and DL algorithms to handle CI problems resulting from this systematic review could cover an entire book on the topic, especially if each algorithm's solution to the CI problem is discussed in more fine details. This is a limitation of the present work and an invitation to expand in a didactic perspective all the material that we found.

The analysis that we did around our second research question, RQ2, highlights a trend in using DM techniques as out-of-the-box solutions, and less awareness about the potential of other ML and DL techniques in addressing the CI problem. Future works should consider applying more sophisticated algorithms, such as the latest DL architectures used for synthetic data generation, because they can substantially increase the quality of the output. On the other hand, further research is needed to explore the limitations and drawbacks, if any, of these new approaches.

While studying the results from RQ3, we noticed an alarming lack of awareness of the importance of using the right metrics, especially in presence of CI. Future works should carefully consider the choice of metrics to use with CI data, and preferably report their results by means of comparisons with other works in the manufacturing domain, over the same set of benchmarking data.

The latest suggestion, it emerges from RQ4, poses another challenge in manufacturing, because there is even a lack of standardized benchmarking datasets to use. We have listed the few public datasets

in the manufacturing domain that are in use at the time of this review; however, future works should try to standardize and regulate the application and comparison of ML solutions in the manufacturing domain.

Finally, we noticed a complete lack of public available repositories and code in manufacturing research. While manufacturing researchers might think that the use of ML in their domain is a mere application of data science research, it should not be neglected the contribution in adapting and testing common data science algorithms to the manufacturing domain. Thus, making publicly available code and benchmarking data should become another standard research practice in manufacturing.

8. Conclusions

In this review article we have addressed the issue of CI in manufacturing. CI poses significant challenges that need to be addressed. Our analysis of 74 articles in literature highlights several key aspects related to this problem.

Firstly, we have identified the most common manufacturing tasks where CI is prevalent. These include defect detection, quality control, anomaly detection, and predictive maintenance, among others. Recognizing these specific areas of imbalance helps researchers and practitioners focus their efforts on developing effective solutions tailored to these manufacturing tasks.

Secondly, we reported how ML and DL techniques, along with DM methods, have been widely adopted to tackle CI in manufacturing. These approaches leverage algorithms that can learn from imbalanced datasets and make accurate predictions. Additionally, DM techniques aim to rebalance the class distribution and improve the performance of classifiers. Understanding the underlying principles of these techniques can guide researchers in selecting appropriate methods for their specific manufacturing contexts. High potential is coming from DL techniques such as CNNs and GANs. Future articles should try to explore more DL solutions, including the most recent transformer architectures.

Thirdly, we assessed and reported how the choice of metrics used to evaluate the performance of algorithms in addressing CI is a crucial consideration. Different manufacturing domains may have specific practices and requirements that influence the selection of appropriate metrics. It is important to carefully consider the trade-offs between metrics such as accuracy, precision, recall, F1-score, and AUC to ensure that the evaluation aligns with the specific goals and constraints of the manufacturing task at hand.

As a result of our review work, we recommend the use of publicly available datasets to facilitate better benchmarking in addressing CI. Standardized datasets enable researchers to compare the performance of different algorithms, establish benchmarks, and track the progress of the field over time. Access to such datasets promotes transparency, facilitates collaboration, and allows for the development of more robust and generalizable solutions.

Furthermore, we emphasize the importance of publicly sharing code repositories. Currently, there is a lack of publicly available code to address CI in manufacturing. Most papers either do not provide code and very few offer it in the form of pseudo-code, limiting reproducibility and hindering the advancement of the field. By openly sharing code, researchers can promote transparency, facilitate replication, and foster a collaborative environment where others can build upon existing work to drive innovation and improve upon established approaches.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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